

SOYAL Protocol 靈活使用技巧

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SOYAL[®]
ACCESS & INDUSTRIAL CONTROL





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25h & 18h 指令差異 / H&E系列 用戶白名單下載差異對照/
AR727CM-IO 實現TCP 轉 可變長度Wiegand 輸出

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③ 常用通訊協議指令實作

Floor Download
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讀卡機 LCD 顯示控制
控制讀卡機 LED & Buzzer 顯示聲音控制
使用語音功能說出 “ 我愛茂旭 ”

1. H系列& E系列通訊協議差異

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所有SOYAL的產品皆使用相同的通訊協定 (Protocol)，對於不同的機型會有不同的控制命令。

1.連線準備：要與卡機通訊前，必須先確認卡機的站號

2.資料格式：

傳輸速率			9600 · N · 8 · 1			
資料格式			Binary HEX 【16進制碼】			
H 系列資料封包						
Head	Length	Node ID	Function Code	Data...	XOR	SUM
8bits	8bits	8bits	8bits		8bits	8bits
7E	XX	0x01	0x18		0xE6	0xFF
起始碼	資料長度	目的站號	功能碼	資料	檢查碼	檢查碼
E 系列資料封包						
Head	Length	Node ID	Function Code	Data...	XOR	SUM
32bits	16bits	8bits	8bits		8bits	8bits
FF.00.5A.A5	XX.XX	0x01	0x18		0xE6	0xFF
起始碼	資料長度	目的站號	功能碼	資料	檢查碼	檢查碼

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	資料長度 (Byte)	功能說明	XOR 0xFF	SUM 0x00
起始碼	1	固定為0x7E	No	No
資料長度	1	本欄位之後，還有幾個有效位元組，包含XOR及SUM	No	No
目的站號	1	所要送達的目的站號。00表示送至電腦，FF表示為廣播	Yes	Yes
功能碼	1	功能碼代表本次傳送的功能。	Yes	Yes
資料	00~DO	資料的位元數依功能碼不同而異，有可能沒有資料組	Yes	Yes
XOR 檢查碼	1	以0xFF與目的站號及其後的位元組做XOR運算		Yes
SUM 檢查碼	1	以0x00與目的站號及其後的位元組做加總運算（包含XOR），若超過0xFF則要取低位元(Low)		

範例: 向站號01查詢訊息

Head	Length	Node ID (DID)	Function Code	Data...	XOR	SUM
0x7E	0x04	0x01	0x18		0xE6	0xFF

$$\text{XOR} = \underbrace{0xFF}_{\text{SUM}} \wedge \underbrace{0x01}_{\text{Node ID (DID)}} \wedge \underbrace{0x18}_{\text{Function Code}} = 0xE6$$

$$\text{SUM} = \underbrace{0x01}_{\text{Node ID (DID)}} + \underbrace{0x18}_{\text{Function Code}} + \underbrace{0xE6}_{\text{XOR}} = 0xFF$$

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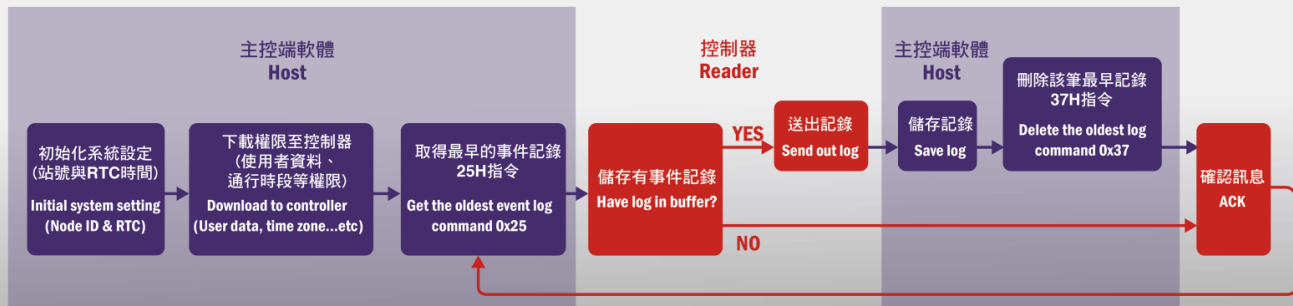
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【25 H】SOYAL標準輪詢運作方式(被動模式)：

控制權在控制端，電腦只查詢歷史事件紀錄



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【 25 H】讀取控制器訊息

	範例	說明
Head	??	Leading Code could be standard format or extended format
Length	04/07	Data length can be one or two bytes depend on protocol format.
Node	01	Node ID of destination ,the node id is 01 here
Command	25	Reading the oldest event log from the device
(Data 1)	(FF)	Extension bytes for specified record read.(Bits 23~16) Note 2.8.1
(Data 2)	(FF)	Extension bytes for specified record read.(Bits 15~18) Note 2.8.1
(Data 3)	(FF)	Extension bytes for specified record read.(Bits 07~00) Note 2.8.1
XOR	DB	$XOR = FF \wedge 01 \wedge 25 = DB$
SUM	01	$SUM + (01 + 25 + DB), FF = 01(LSB)$

訊息檔案的資料結構:

701Server程式由控制器取得訊息後，以訊息資料結構存於以該訊息發生日期為檔名的檔案中。

每一筆訊息固定存成100Bytes。

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Command Echo Format

After command executed the target controller will echo result code and append data on followed format.

Echo	Value	Description
Head	??	The header will be same with send command. (0x7E or 0xFF005AA5)
Length	00 ??	Data Length bytes will be same with send command. (1 or 2 bytes)
Node	00	Always 00, mean this is one echo package to host where command from.
Function Echo		Command Echo Code , Reference to Echo Table
Source ID	01	The source node ID of data package come from (Who send)
Data 0~n		Followed data will be appended after 2009.FEB.09
Check Sum	81	Appended Data Field , This field will depend on different controller.

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Has Event: Echo the oldest Log

- Function code define table

00	Site code error
01	Invalid user PIN
02	Keypad Locked by error access over limits times
03	Invalid card
04	Time Zone error
05	Door Group error
06	Expiry Date
07	Over access times
08	PIN Code error
09	Press duress PB
10	Access by Card and PIN

11	Normal Access by tag
12	Force Controller Relay ON
13	Force Controller Relay Off
14	Controller armed
15	Controller disarmed
16	Egress
17	Alarm event
20	Controller Power Off
21	Duress
22	Guards for help
23	Cleaner access



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Has Event: Echo the oldest Log

- Function code define table

24	Controller Power On
25	Force Controller Relay On/Off Error
26	Reader Return to Normal(RTN)
27	Help push button pressed
28	Access by PIN (Key Only)
29	Digital input actives
30	RS485 slave reader off line (Disconnected)
31	RS485 slave reader on line (Connected)
32	User PIN code changed
33	Change user PIN error
34	Controller Power On

35	Enter Auto Door Open Procedure
36	Exit Auto Door Open Procedure
37	Auto Disarmed
38	Auto Armed
39	Access by fingerprint or finger vein
40	Fingerprint identify failed
41	
42	Remote control "Up Key" pressed
43	Disable Reader
44	Enable Reader
45	Remote control "Panic Key" pressed



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Has Event: Echo the oldest Log

- Function code define table

46	User entrance at parking system
47	User exit at parking system
48	Counter triggered at parking system
49	Latch Relay
50	
51	
52	
53	Enter/Exit Edit Mode
54	
55	Free Access Mode Enable/Disable (SubFunc 0/1 is Dis/Enable)在721Ev2底下的卡機 直接設定『見卡即開』·開放全部卡片通行
56	Access via Fingerprint Error

57	
58	
59	Inhibit card while door open
60	Never open door after card accessed
61	
62	Can' t Read Date/Time from mifare card
63	Can' t Read command from mifare card
64	Can' t deduct from mifare card
65	SOR global card accessed
66	SOR disturber layer error
67	Access rejected : before begin date

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Has Event: Echo the oldest Log

- Function code define table

68	Access rejected : expiry	80	
69	Access rejected : card value not enough	81	
70	Access ok and card value deducted	82	
71	Access ok and read lift data from media failed	83	
72	SOR global access ok and card value deducted	84	
73	SOR global access reject by value not enough	85	
74	SOR global access ok but card value deducted failed	86	Black table tag accessed
75	SOR global access ok without deducted	87	
76		88	
77		89	
78		90	
79			
		100	Access ok : access via vein
		101	Access reject : access via vein
		102	Inhibited by internal lock locked for door Lock
		103	
		104	Fire alarm input triggered

Log format and log Code

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【25 H】Data format of event log (企業版控制器)

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	21	Data length can be one or two bytes depend on protocol format.
Node ID	00	Echo to host
Function code	27	Event function code. Please reference to "3.2 Function code define table"
Data 0	??	The source node ID of data package come from (Who send)
Data 1	??	Second
Data 2	??	Minute
Data 3	??	Hour
Data 4	??	Weekday
Data 5	??	Day
Data 6	??	Month
Data 7	??	Year (Range 00~99, that is mean 2000~2099)
Data 8	??	Port Number. 17:Main port, 18:Port WG1. 19: Port WG2

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【25 H】Data format of event log (企業版控制器)

	Value	Description
Data 9	??	User address bit 15~08 of normal access. (Tag ID bit 15~08 of invalid card))
Data 10	21	User address bit 07~00 of normal access. (Tag ID bit 07~00 of invalid card)
Data 11	Sub Code	Sub message code
Data 12	Sub Func.	Set bit 7 for force open in alarm event.(function code: 17)
Data 13	Ext Code	Option byte of event port for normal access.
Data 14	User level	Bit 5~0 are user level. Set bit 6 for free access event. Set bit7 for multi door controller or set for WG port event at single door controller.
Data 15	Tag ID	Tag ID bit 31~24 of normal access.
Data 16	Tag ID	Tag ID bit 23~16 of normal access.
Data 17	Door Num.	The door number of event port.
Data 18	??	
Data 19	Tag ID	Tag ID bit 15~08 of normal access.
Data 20	Tag ID	Tag ID bit 07~00 of normal access.
Data 21	Value/Tag ID	SOR: Tag deduction amount. Bit 47~40 of 8 Byte UID

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【25 H】Data format of event log (企業版控制器)

	Value	Description	
		(Bit 15~08 of deduction access)	
Data 22	Value/Tag ID	SOR: Tag deduction amount. (Bit 07~00 of deduction access)	Bit 39~32 of 8 Byte UID
Data 23	Balance/ Tag ID	SOR: Tag balance. (Bit 15~08 of deduction access)	Bit 63~56 of 8 Byte UID
Data 24	Balance/ Tag ID	SOR: Tag balance. (Bit 07~00 of deduction access)	Bit 48~55 of 8 Byte UID
Data 25	27	User Inputted Code (Bit 31~24) Note1, (Only valid for event code 0x01)	
Data 26	??	User Inputted Code (Bit 23~16) Note1 , (Only valid for event code 0x01)	
Data 27	??	User Inputted Code (Bit 15~8) Note1 , (Only valid for event code 0x01)	
Data 28	??	User Inputted Code (Bit 0~7) Note1 , (Only valid for event code 0x01)	
XOR		XOR	
SUM		SUM	

Echo Case of Polling(25H)

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**【Case 1】No Event: Echo Current Status**

• Echo Code List

Echo Code	Description	Note
03h	Echo requested data	7E xx 00 03 [Data...] checksum
04h	Command acknowledged (ACK)	7E 04 00 04 FB FF
05h	Command unacknowledged (NACK)	7E 04 00 05 FA FF
06h	Authentication failed (AUTHERR)	7E 04 00 06 F9 FF
07h	Tag operation error: no tags presented (NOTAG)	7E 04 00 07 F8 FF
08h	Tag operation error: User not login (NOT LOGIN LEVEL2)	7E 04 00 08 F7 FF
09h	Tag operation error: CRC8 check error for ReadBlockCRC8	7E 04 00 09 XX XX
0Ah	Tag operation error: Data sector not authenticated	
0Bh	Tag operation error: Data sector authentication error	
0Ch	Reject command for communication level error.	
0Dh	TCP Linker Operation Timeout	

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【Case 2】Invalid Card Event Log Data Fileds

Echo	Value	Description	
Head	7E	Leading Code	
Length	1D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Event	03	Invalid Cards	
Reader ID	01	Reader ID	
Data Field	01	Data 0	Second
	35	Data 1	Minute
	01	Data 2	Hour
	02	Data 3	Day of a week
	05	Data 4	Date
	06	Data 5	Month
	06	Data 6	Year
	01	Data 7	Represent the message source ID

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【Case 2】Invalid Card Event Log Data Fileds

Data Field	04	Data 8	Addr Hi	Addr H Addr L=0x400 (01024) User Address : 1024
	00	Data 9	Addr Lo	
	00	Data 10	On Duty	
	00	Data 11	Bit7 : Forced Open Alarm	
	18	Data 12	Bit Selection (20*xxx#)	
	00	Data 13	Represent the message source comes from WG, bit7 sets 1.	
	63	Data 14	Site Hi	Tag ID bit 31~16 = 0x636B
	6B	Data 15	Site Lo	
	01	Data 16	Door Number (Same as Node)	
	14	Data 17	401RO16 Parameter Setting (24*xxx#)	
	B8	Data 18	Card Hi	Tag ID bit15~0 = 0xB8B4
	B4	Data 19	Card Lo	



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【Case 2】Invalid Card Event Log Data Fileds

If SOR money deduct off, then this 4 bytes are meaningless	00	Data 20	Deducted amount
	00	Data 21	
	00	Data 22	Balance
	00	Data 23	
XOR	C3	XOR=FF^00^03^01^01^35^01^02^05^06^06^01^04^00^00^00^18^00^63^6B^01^14^B8^B4^00^00^00^00=C3	
SUM	7D	SUM=(00+03+01+01+35+01+02+05+06+06+01+04+00+00+00+18+00+63+6B+01+14+00+00+00+00+C3) · FF=7D(LSB)	

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【Case 2】Vaild Card Event Log Data Files

Echo	Value	Description	
Head	7E	Leading Code	
Length	1D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Event	0B	valid Cards	
Reader ID	01	Reader ID	
Data Field	36	Data 0	Second
	21	Data 1	Minute
	02	Data 2	Hour
	02	Data 3	Day of a week
	05	Data 4	Date
	06	Data 5	Month
	06	Data 6	Year
	01	Data 7	Represent the message source ID

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【Case 2】Vaild Card Event Log Data Files

Data Field	00	Data 8	Addr H	Addr H Addr L=0x01(00001) User Address : 00001
	01	Data 9	Addr L	
	00	Data 10	On Duty	
	00	Data 11	Bit7 : Forced Open Alarm	
	18	Data 12	Bit Selection (20*xxx#)	
	00	Data 13	Represent the message source comes from WG, bit7 sets 1.	
	63	Data 14	Site Hi	Site Hi Site Lo = 0x636B (25451)
	6B	Data 15	Site Lo	
	01	Data 16	Door Number (Same as Node)	
	14	Data 17	401RO16 Parameter Setting (24*xxx#)	
	B8	Data 18	Card Hi	Card Hi Card Lo = 0XB8B4 (27284)
	B4	Data 19	Card Lo	



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【Case 2】Vaild Card Event Log Data Files

If SOR money deduct off, then this 4 bytes are meaningless	00	Data 20	Deducted amount
	00	Data 21	
	00	Data 22	Balance
	00	Data 23	
XOR	EE	XOR=FF^00^0B^01^36^21^02^02^05^06^06^01^00^01^00^00^18^00^63^6B^01^14^B8^B4^00^00^00^00=EE	
SUM	CF	SUM=(00+0B+01+36+21+02+02+05+06+06+01+00+01+00+00+18+00+63+6B+01+14+B8+B4+00+00+00+00+EE) · FF=CF	

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No Events

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Invalid Cards

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Valid Card

事件案例介紹

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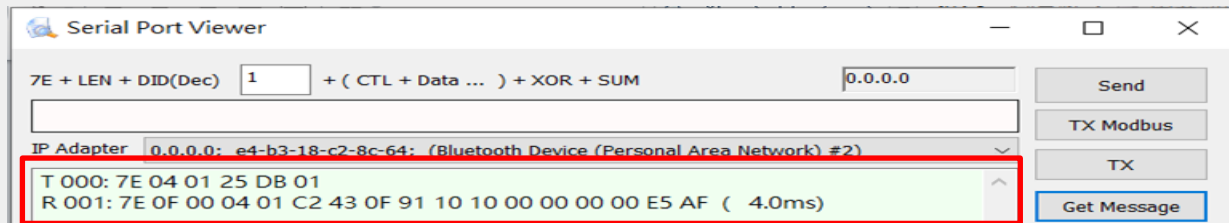
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No Events

此範例以AR-837-E 控制器為例。



T 000: 7E 04 01 25 DB 01

R 001: 7E 0F 00 04 01 C2 43 0F 91 10 10 00 00 00 00 E5 AF (4.0ms)

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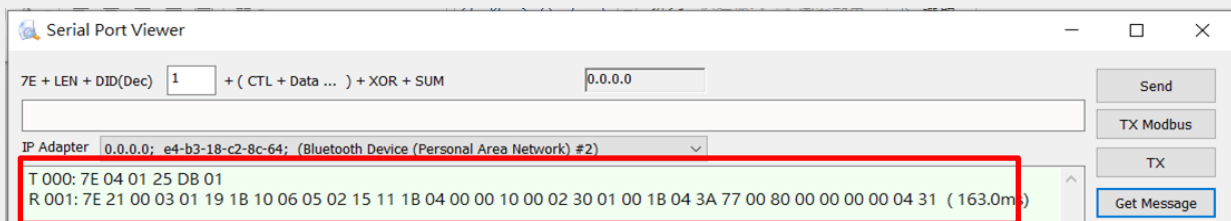
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Invalid Cards

此範例以AR-837-E 控制器為例。



T 000: 7E 04 01 25 DB 01

R 001: 7E 21 00 **03** 01 19 1B 10 06 05 02 15 11 1B 04 00 00 10 00 02 30 01 00 1B 04 3A 77
00 80 00 00 00 00 04 31 (163.0ms)

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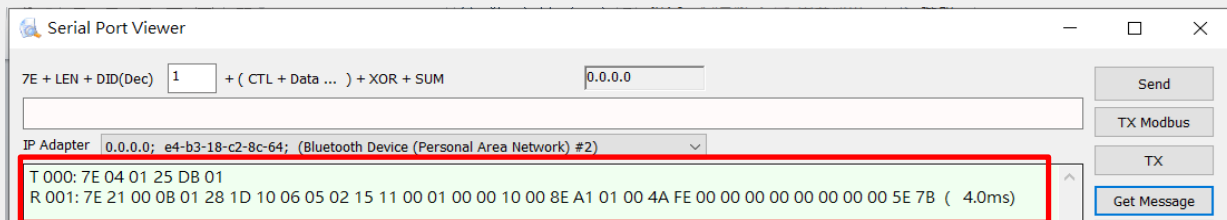
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Valid Card

此範例以AR-837-E 控制器為例。



T 000: 7E 04 01 25 DB 01

R 001: 7E 21 00 0B 01 28 1D 10 06 05 02 15 11 00 01 00 00 10 00 8E A1 01 00 4A FE
00 00 00 00 00 00 00 00 5E 7B (4.0ms)

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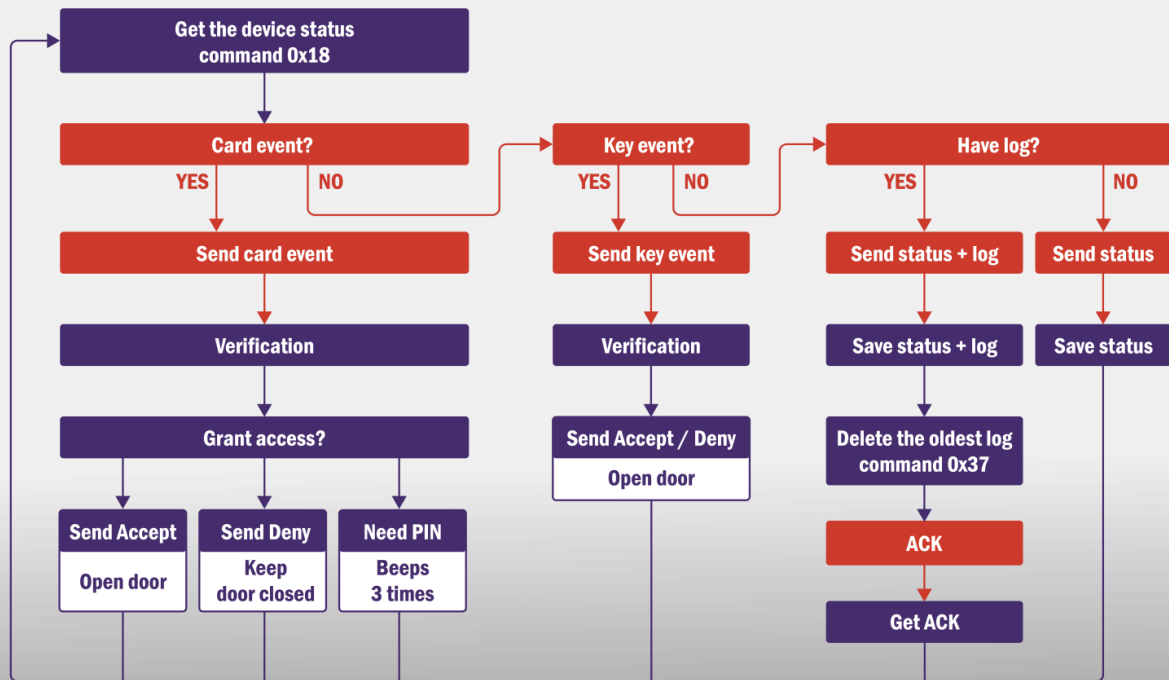
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【 18 H】SOYAL主控模式：

主控權在電腦端，即時查詢卡機每一個動作行為並即時回應



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【 18 H】 Hosting Polling

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	04	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination
Command	18	Hosting Polling
	*	[Second], Optional data
	*	[Minute]
	*	[Hour]
	*	[Weekday]
	*	[Day]
	*	[Month]
	*	0x00
	*	[Weekday]
	*	[Year]
XOR	E6	$XOR = FF \oplus 01 \oplus 18 = E6$
SUM	FF	$SUM = 01 + 18 + E6 = FF$

P.S.: The optional data can support real time clock synchronization between slave and host.

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



【 18 H】Hosting Polling

Echo:

1. The device respond data will depending on different status as follows
2. Standby status w/o event log
3. User key buffer status for PIN only or card and PIN.
4. Card flashing Status
5. Standby status with event log
6. Echo hosting keypad buffer if keypad is in hosting mode

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



【 18 H】Hosting Polling

	Value	Description
Head	7E	Leading Code
Length	??	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Destination	00	Destination Node ID. 00H mean send to host (PC or AR716E)
Function Code	09	09H is echo reader status code.
Source	01	Source Node ID. (Who send);
Event	×	Type of message echo from device (please refer Event Table)
Data Field	Data 0	Data 0
XOR		Check byte
SUM		Check byte

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



【 18 H】Hosting Polling

Event Table:

Event	Meaning	Description of Data Bit
00	I/O status	Neither card nor key event, the controller will echo device I/O status. Data 0: I/O Status Byte: Bit7: Keypad status. If keypad locked by errors than this bit will be set Bit6: Door release output status of main port. (1/0: On/Off) Bit5: Alarm output Status. Bit4: Arming status. Bit3: Alarm status of controller. Bit2: RFU Bit1: Egress Button status. (1/0: Release/Pressed) Bit0: Door sensor status. (1/0: Open/Close) Data 1: Bit7: Force alarm status. (1/0: Force alarm / Open too long alarm), only available in Data 0 bit5 is set. Data 2: Controller' s Option Parameters (Data 32 or 33 for command 0x12, sub code 0x00) Bit7: Anti-pass back (1/0: Ena/Dis) Bit6: In/Out for anti-pass back (1/0: In/Out) Bit5: Force Open Alarm (1/0:Enable/Disable) Bit4: Egress Button (1/0:Enable/Disable) Bit3: Skip PIN check while Card & PIN access mode. Bit2: Auto Open Zone (1/0:Enable/Disable) Bit1: Auto lock door at door closed (1/0:Enable/Disable) Bit0: Is time attendance door (1:0: No/Yes) Data 3: RFU. Always 0.

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



【 18 H】Hosting Polling

Event Table:

Event	Meaning	Description of Data Bit
01	4/5 Keys pressed Note : Mode4: 5 keys pressed, Mode8: 4 keys pressed	Data 0 : The 5th Key data. Set Bit7 while working on Mode 8. Data 1 : Input value' s MSB Data 2 : Input value' s LSB Data 3 : Undefined default value=0 Data 4 : Device parameters. (Setting by command: 20 * XXX #) Data 5 : 401RO16' s parameter (24*xxx#) Data 6 : Undefined default value=0 Controllers are allow to identify the value Data 7~ Data 11 : M4 : The Key data M8 : Data 7~Data 10 = Key data , Data 11 = 00 Data 12 : Undefined default value=0

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



【 18 H】Hosting Polling

Event Table:

Event	Meaning	Description of Data Bit																									
02	New Card Present UID4: Tag ID Bits(39~32) UID3: Tag ID Bits(31~24) UID2: Tag ID Bits(23~16) UID1: Tag ID Bits(15~08) UID0: Tag ID Bits(07~00)	The inner code of the card is 40bits in length which can be tabled 7E xx 00 09 01 02 Dat0 UID3 UID2 Dat3 Dat4 UID1 UID0 UID4..... XOR SUM below: <table><tr><th>MSB</th><th colspan="2">Inner Code</th><th colspan="2">LSB</th></tr><tr><th>ID</th><th colspan="2">Site Code</th><th colspan="2">Card Code</th></tr><tr><td></td><th>High</th><th>Low</th><th>High</th><th>Low</th></tr><tr><th>ID</th><th>Site Hi</th><th>Site Lo</th><th>Card Hi</th><th>Card Lo</th></tr><tr><td>8 Bits 39~32</td><td>8 Bits 31~24</td><td>8 Bits 23~16</td><td>8 Bits 15~08</td><td>8 Bits 07~00</td></tr></table>	MSB	Inner Code		LSB		ID	Site Code		Card Code			High	Low	High	Low	ID	Site Hi	Site Lo	Card Hi	Card Lo	8 Bits 39~32	8 Bits 31~24	8 Bits 23~16	8 Bits 15~08	8 Bits 07~00
MSB	Inner Code		LSB																								
ID	Site Code		Card Code																								
	High	Low	High	Low																							
ID	Site Hi	Site Lo	Card Hi	Card Lo																							
8 Bits 39~32	8 Bits 31~24	8 Bits 23~16	8 Bits 15~08	8 Bits 07~00																							



25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



【 18 H】Hosting Polling

Event Table:

Event	Meaning	Description of Data Bit
02	New Card Present UID4: Tag ID Bits(39~32) UID3: Tag ID Bits(31~24) UID2: Tag ID Bits(23~16) UID1: Tag ID Bits(15~08) UID0: Tag ID Bits(07~00)	Data 0 : Time & Attendance · Bit7~Bit5: Time & Attendance Selection Other LSBs refer to "Exit Input" Data 1 : High byte of site code (bit 15~08) Data 2 : Low byte of site code (bit 07~00) Data 3/4 : The MSB / LSBs of the value input before card flashing, it would show the previous value if there is not value input before flashing Data 5 : High byte of card code (bit 15~08) Data 6 : Low byte of card code (bit 07~00) Data 7 : ID Code(Bits(39~32) of EM4001 Chi Data 8 : Bit Selection (20*xxx#) Data 9 : Bit7:1/0 Set for multi door Controller or WG port event at single door controller. Bit6: 1/0 Setting Forced Open Alarm

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



【 18 H】Hosting Polling

Event Table:

Event	Meaning	Description of Data Bit
06	Keypad buffer	Echo Keypad buffer status in hosting mode. The followed data we be: Data 1: fixed to 0x80 Data 2: Key counter (n), how many key had pressed. Data 3 ~ 3+n: Key data.

Invalid Tag examples

Host Polling :7E 0D 01 18 20 2B 0B 08 04 00 01 12 00 F9 87

Slave Echo :7E 10 00 09 01 02 0B 10 01 00 00 10 01 77 C8 00 41 B9

Host Responce:7E 0B 01 05 00 10 01 3A 98 10 01 59 53

Card Only Tag example:

Host Polling :7E 0D 01 18 0D 2C 0B 08 04 00 01 12 00 D3 4F

Slave Echo :7E 10 00 09 01 02 0B 00 65 00 00 0F C5 42 C8 00 DB 35

Host Responce:7E 0D 01 04 00 0F C5 00 4E 00 00 00 65 1B A7

Card and PIN Tag example:

Host Polling :7E 0D 01 18 17 3A 0B 08 04 00 01 12 00 DF 73

Slave Echo :7E 10 00 09 01 02 0B 04 D5 00 00 B8 26 01 C8 00 78 0F

Host Responce:7E 0D 01 09 04 B8 26 00 59 16 2E 04 D5 99 37

Host polling :7E 0D 02 18 17 3A 0B 08 04 00 01 12 00 DC 71 (Polling next Node ...)

....

Host polling :7E 0D 01 18 1B 3A 0B 08 04 00 01 12 00 D3 6B (The Node who got Tag)

Slave Echo :7E 18 00 09 01 03 0B 00 59 02 C8 16 2E 05 06 07 08 0B 00 E7 A7 8D 8C 7F 6D 37 (User Input)

Host Responce:7E 0D 01 04 08 B8 26 00 59 00 00 04 D5 E4 01

Host polling :7E 0D 02 18 1B 3A 0B 08 04 00 01 12 00 D0 69 (Polling next Device)

;This tag is Card & PIN

» H系列 & E系列通訊協議差異

» 常用通訊協議指令概說

» 常用通訊協議指令實作

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

18H資料格式
說明

18H事件說明



Standby status w/o event log

操作說明

User key buffer status for PIN
only or card and PIN.

操作說明

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

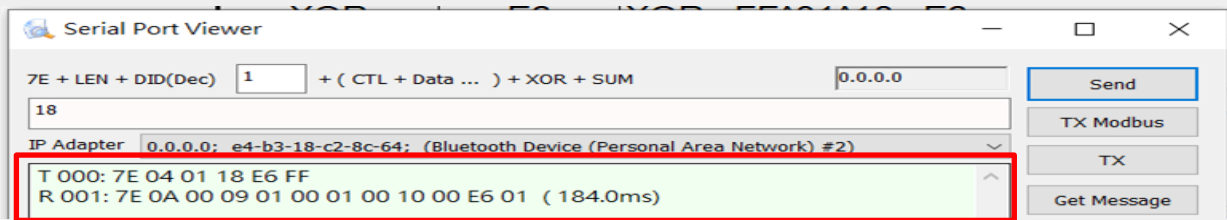
18H資料格式
說明

18H事件說明



Standby status w/o event log

此範例以AR-837-E 控制器為例。



T 000: 7E 04 01 18 E6 FF

R 001: 7E 0A 00 09 01 00 01 00 10 00 E6 01 (184.0ms)

(09H is echo reader status code.)

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

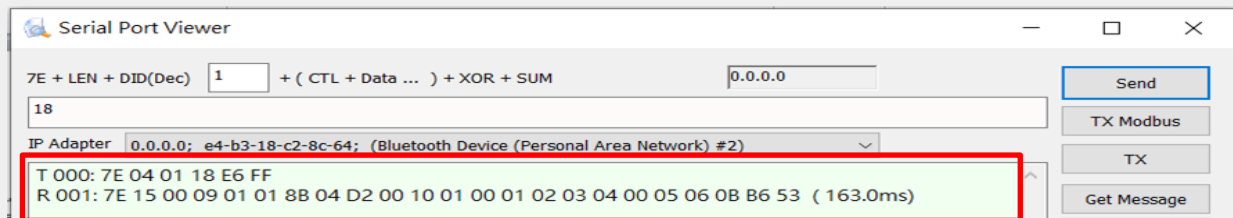
18H資料格式
說明

18H事件說明



User key buffer status for PIN only or card and PIN

此範例以AR-837-E 控制器為例。



- 卡機操作：按鍵上按數字1 2 3 4

T 000: 7E 04 01 18 E6 FF

R 001: 7E 15 00 09 01 01 8B 04 D2 00 10 01 00 01 02 03 04 00 05 06 0B B6 53 (163.0ms)

25/18通訊協定
資料格式

25/18資料封包

25H資料格式
說明

25H事件說明

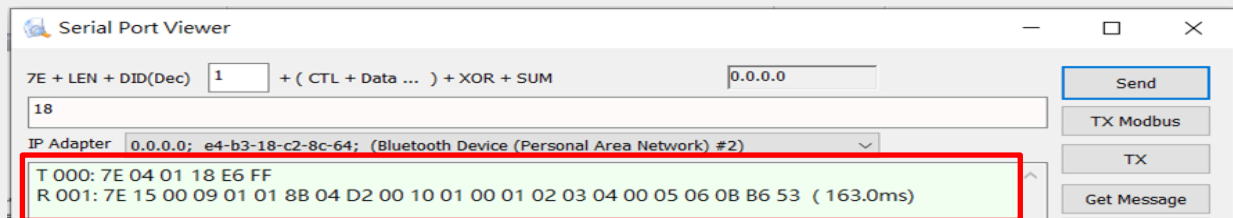
18H資料格式
說明

18H事件說明



User key buffer status for PIN only or card and PIN

此範例以AR-837-E 控制器為例。



- 卡機操作:感應一張卡片取得的訊息-----卡號:Hex 02 30 1B 04

T 000: 7E 04 01 18 E6 FF

R 001: 7E 17 00 09 01 02 0B 02 30 00 00 1B 04 77 10 00 8A 05 00 00 80 3A 77 F6 A5
(38.0ms)

H&E系列用戶白名單下載差異對照

H系列控制器/E系列控制器 對照表

Echo	E Series Value	H Series Value	Description
Head	7E	7E	Leading Code could be standard format or extended format
Length	1F	0E	Data length can be one or two bytes depend on protocol format.
DID	01	01	Node ID of destination, the node id is 01 here
CMD	83	83	83H:with anti-passback , 84H:without anti-passback flag byte
Records	01		How many records of user will be download to
User Addr	00	00	Two bytes of user address. (Little endian)
User Addr	01	01	Two bytes of user address. (Little endian)
Tag UID	00		Eight bytes of Tag ID. (Little endian)
	00		
	00		
	00		
	8E	8E	
	A1	A1	
	4A	4A	
	FE	FE	

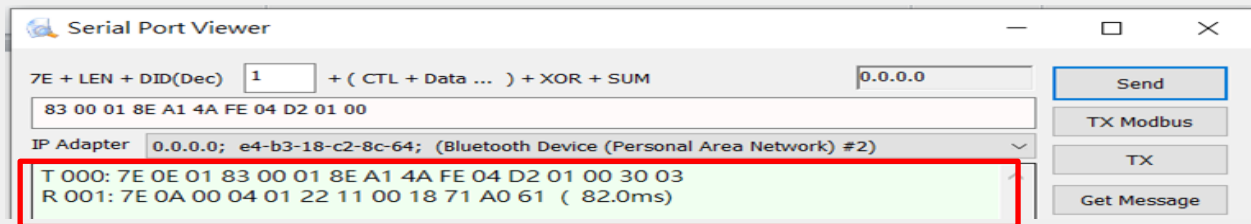
H系列控制器/E系列控制器 對照表

Echo	E Series Value	H Series Value	Description
PIN Code	00		Four bytes of PIN code. (Bit 31 ~ 24) ※3 Bitmask for HTZ 9~16
	00		Four bytes of PIN code. (Bit 23 ~ 16) ※3 Bitmask for HTZ 1 ~ 8
	00	04	Four bytes of PIN code. (Bit 15 ~ 8) Word PIN Code (16 Bits)
	00	D2	Four bytes of PIN code. (Bit 7 ~ 0)
Mode	58	01	Bit7/6: Access Mode Bit5 : patrol card Bit4 : Card omitted after fingerprint recognition. Bit3 : Fingerprint omitted after card recognition. Bit2 : enable expire check Bit1 : enable anti-pass back control Bit0 : password change available
			Access Mode Define (Bit 7/6)
			00 Invalid
			01 Read Only
			10 Card or PIN
			11 Card + PIN
Zone	00	00	Bit 7: Multi door controller: Set to enable independent time zone for each node. (Cmd: 89H) Single door controller: Set to 0 if WG port will use the same time zone. Otherwise, the WG port will accord to Zone2. Bit 6: Reserved, must be 0. Bit5~0: User Access Time Zone. Zero for free zone control

H系列控制器/E系列控制器 對照表

Echo	E Series Value	H Series Value	Description
Group 1	FF		Bit mapping of available door. (Door 16~9/Bit 7~0) ※1
Group 2	FF		Bit mapping of available door. (Door 16~9/Bit 7~0) ※1
Year	00		Year of the last allowed date at 2 digital format
Month	00		Month of the last allowed date at 2 digital format
Day	00		Day of the last allowed date at 2 digital format
Level	00		Bit7~6: User Level define range 0x00 to 0x03 Bit5~0: Zone 2 for WG port.
Option	00		Extend option Byte. Bit7: 1/0 En/Disable Anti-pass-back check Only available for cmd:83H, for 84H this byte will be discarded.
Reserved	00		
	00		
	00		
N*26	...		Followed the next record of user data if records more than one set
XOR	BE	30	$XOR = FF \wedge DID \wedge CMD \wedge \dots \wedge \dots \wedge Level$
SUM	11	03	$SUM = (DID + CMD + \dots + \dots + Level + XOR)$

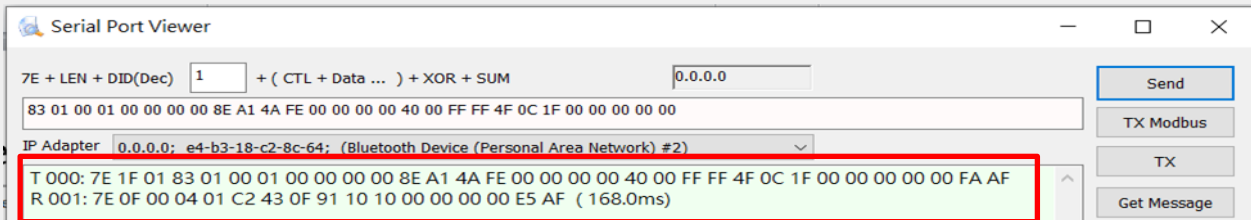
H系列控制器



T 000: 7E 0E 01 83 00 01 8E A1 4A FE 04 D2 01 00 30 03

R 001: 7E 0A 00 04 01 22 11 00 18 71 A0 61 (82.0ms)

E系列控制器



T 000: 7E 1F 01 83 01 00 01 00 00 00 00 8E A1 4A FE 00 00 00 00 40 00 FF FF 4F 0C
1F 00 00 00 00 00 FA AF

R 001: 7E 0F 00 04 01 C2 43 0F 91 10 10 00 00 00 00 E5 AF (168.0ms)

AR-727CM-IO 實現TCP 轉 可變長度Wiegand 輸出

» H系列 & E系列通訊協議差異

» 常用通訊協議指令概說

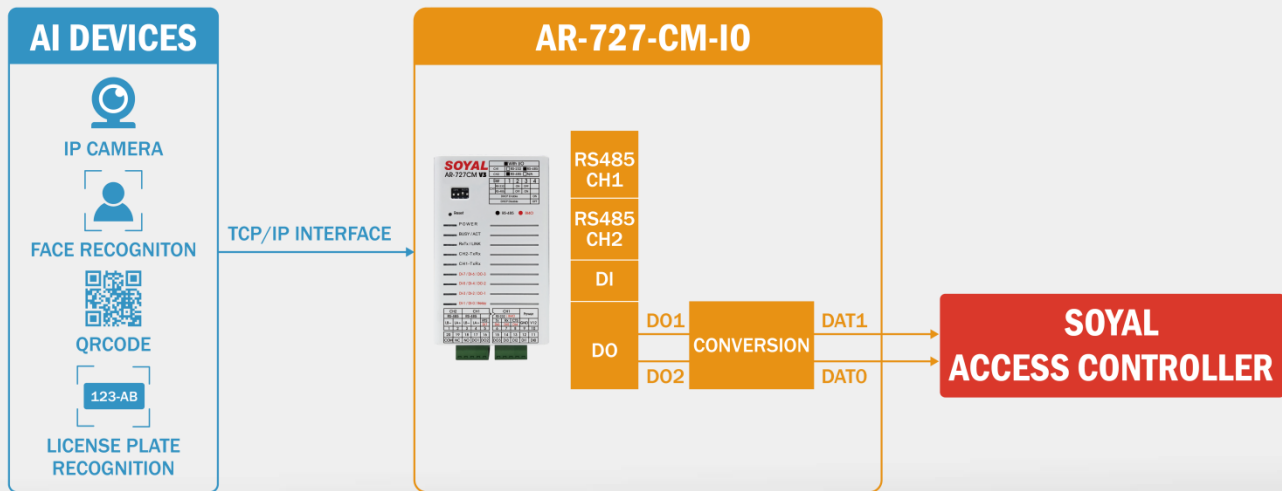
» 常用通訊協議指令實作

案例介紹

產品架構/說明

操作驗證

示意圖:



案例介紹

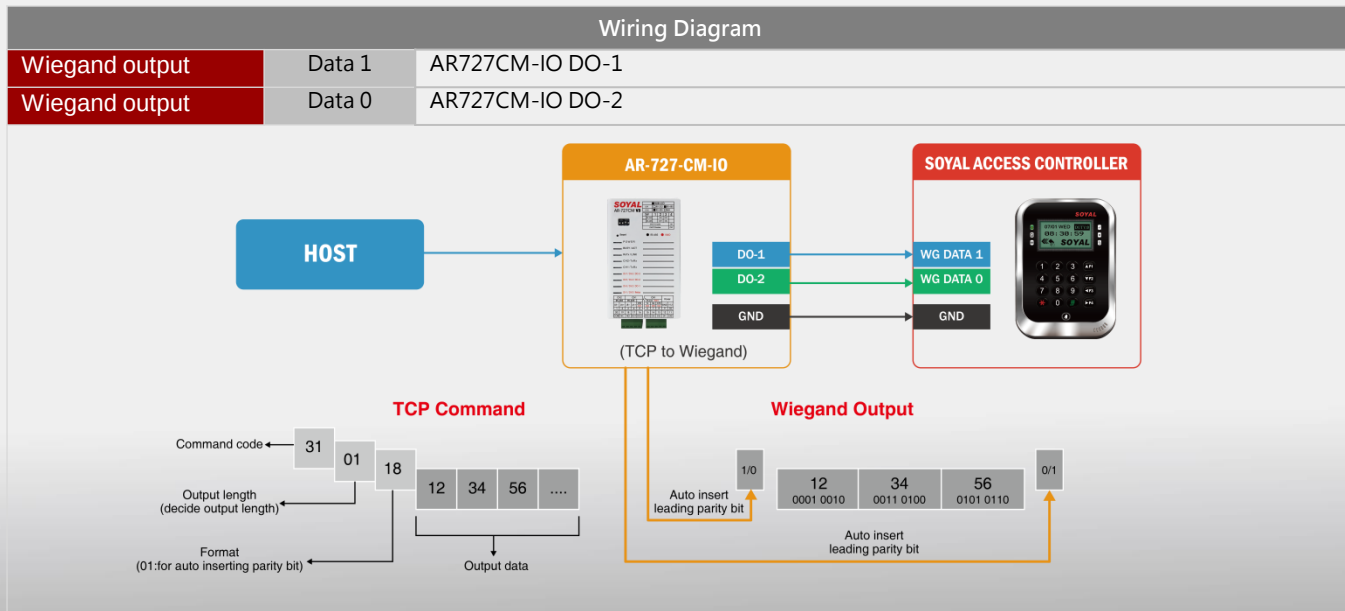
產品架構/說明

操作驗證

IMPLEMENTATION STEPS :

韌體版本: APX727i3__V0500 nologo 8i4o 191014 Wiegand Output.STM

Wiegand output with changeable data length (Simple format): 0x31



案例介紹



產品架構/說明



操作驗證



IMPLEMENTATION STEPS :

韌體版本: APX727i3__V0500 nologo 8i4o 191014 Wiegand Output.STM

Format Explanation

TCP Port	1601	Output data bit length is changeable depending on demand, supporting 16~96 bits
Command code	0x31	Wiegand output command Output Format: 0x00: Just bypass the data bits through WG1/0 pin 0x01: Auto insert leading/ending parity bit into the bypass data bits on even/odd format 0x02: Auto insert leading/ending parity bit into the bypass data bits on odd/even format

Example

Command Package					
Command	Format	Auto add parity bit	Data bit length (changeable, support 16~96 bits)	Data bit (hex)	Pass output data bits (bin)
0x31	0x00	None	0x20 / 32dec (actually send 0x20 / 32bits)	12 34 56 78	0001 0010 0011 0100 0101 0110 0111 1000
0x31	0x01	Even / Odd	0x20 / 32dec (actually send 0x22 / 34bits)	12 34 56 78	<u>1</u> 0001 0010 0011 0100 0101 0110 0111 1000 1
0x31	0x01	Even / Odd	0x18 / 24dec (actually send 0x1A / 26bits)	12 34 56	<u>0</u> 0001 0010 0011 0100 0101 0110 0
0x31	0x02	Odd / Even	0x18 / 24dec (actually send 0x1A / 26bits)	12 34 56	<u>1</u> 0001 0010 0011 0100 0101 0110 1

案例介紹



產品架構/說明



操作驗證



IMPLEMENTATION STEPS :

韌體版本: APX727i3__V0500 nologo 8i4o 191014 Wiegand Output.STM

Echo code		
	0x40	COMMAND_RET_SUCCESS
	0x41	COMMAND_RET_UNKNOWN_CMD
	0x42	COMMAND_RET_INVALID_CMD
	0x43	COMMAND_RET_INVALID_ADR

More Examples	
AR-727-CM sends 96 bits data with parity bit format: 31 01 60 10 11 22 33 44 55 66 77 88 99 AA BB	
Pass output data bits: 1 0001 0000 0001 0001 0010 0010 0011 0011 0100 0100 0101 0101 0110 0110 0111 0111 1000 1000 1001 1001 1010 1010 1011 1011 0	

案例介紹

測試工具可以用  Packet Sender

來驗證送出的值是否有成功，如下圖所示：

產品架構/說明

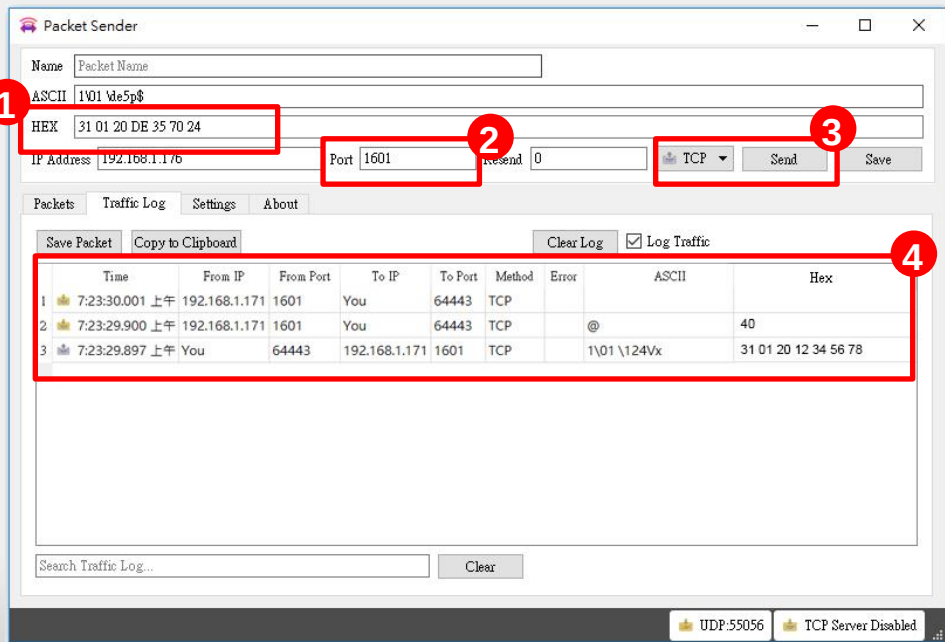
操作驗證

Step 1 :
設定IP/送出的數值

Step 2 : Port: 1601

Step 3 :
選擇 TCP Port 再按Send

Step 4 :成功後會有回應值



2.常用通訊協議指令概說

» H系列& E系列通訊協議差異

» 常用通訊協議指令概說

» 常用通訊協議指令實作



25H

Get the oldest event log
of device
(讀取控制器訊息)



37H

Remove the oldest
event log of device
(移除控制器訊息)



83H/84H

Set User Parameters
(新增卡片)



87H

Get User Parameters
(確認卡片是否存在)



85H

Erase user data
(刪除卡片)



21H

Relay On/Off Control
(開門)

【25H】 / 【37】 流程圖介紹





【25H】 Get the oldest event log of device

	範例	說明
Head	??	Leading Code could be standard format or extended format
Length	04/07	Data length can be one or two bytes depend on protocol format.
Node	01	Node ID of destination ,the node id is 01 here
Command	25	Reading the oldest event log from the device
(Data 1)	(FF)	Extension bytes for specified record read.(Bits 23~16) Note 2.8.1
(Data 2)	(FF)	Extension bytes for specified record read.(Bits 15~18) Note 2.8.1
(Data 3)	(FF)	Extension bytes for specified record read.(Bits 07~00) Note 2.8.1
XOR	DB	$XOR = FF \wedge 01 \wedge 25 = DB$
SUM	01	$SUM + (01 + 25 + DB), FF = 01(LSB)$



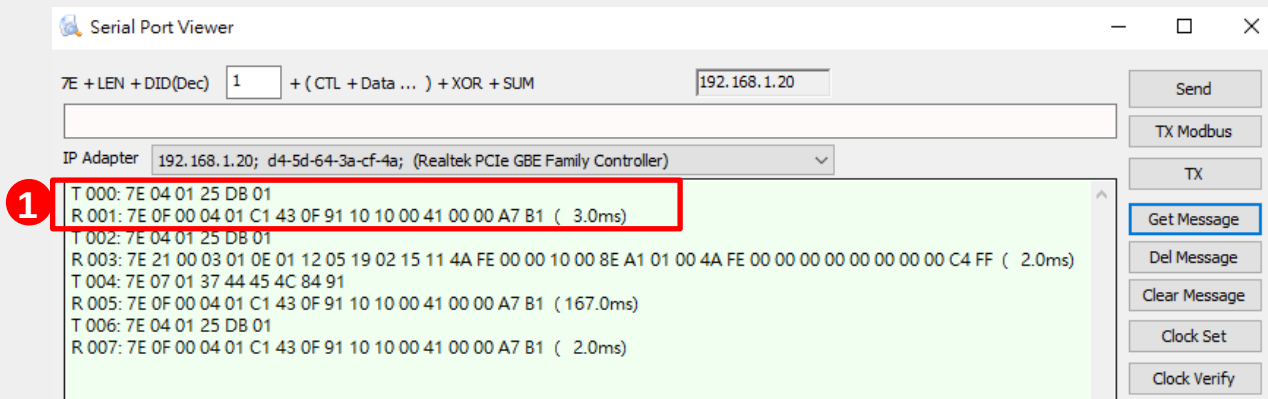
【37H】 Remove the oldest event log of device

	Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Node ID	01	Node ID of destination, the node id is 01 here
Command	37	Remove the lasted log
XOR	C9	$\text{XOR} = \text{FF} \wedge 01 \wedge 37 = \text{C9}$
SUM	01	$\text{SUM} = (01 + 37 + \text{C9}) \cdot \text{FF} = 01(\text{LSB})$
Head	7E	Leading Code



【25H】 / 【37H】

STEP 1 :查詢卡機是否有歷史記錄存在



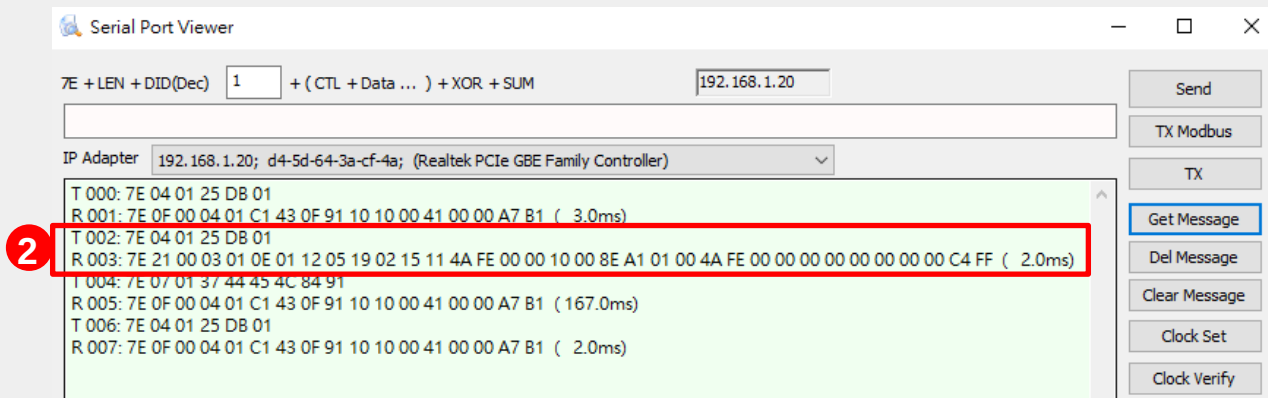
T 000: 7E 04 01 25 DB 01

R 001: 7E 0F 00 04 01 C1 43 0F 91 10 10 00 41 00 00 A7 B1 (3.0ms)



【25H】 / 【37H】

STEP 2 : 刷卡後，再次查詢卡機



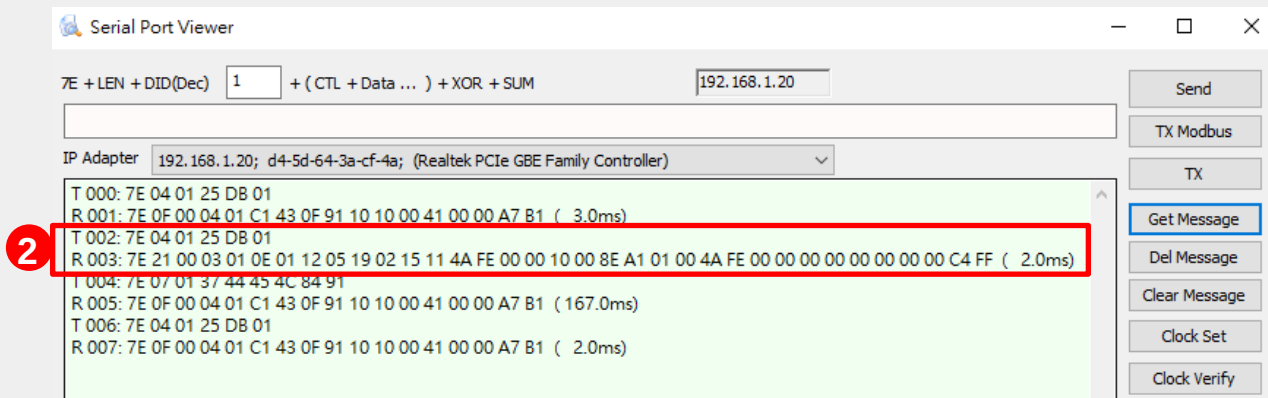
T 002: 7E 04 01 25 DB 01

R 003: 7E 21 00 03 01 0E 01 12 05 19 02 15 11 4A FE 00 00 10 00 8E A1 01 00 4A FE
00 00 00 00 00 00 00 00 C4 FF (2.0ms)



【25H】 / 【37H】

STEP 2 : 刷卡後，再次查詢卡機



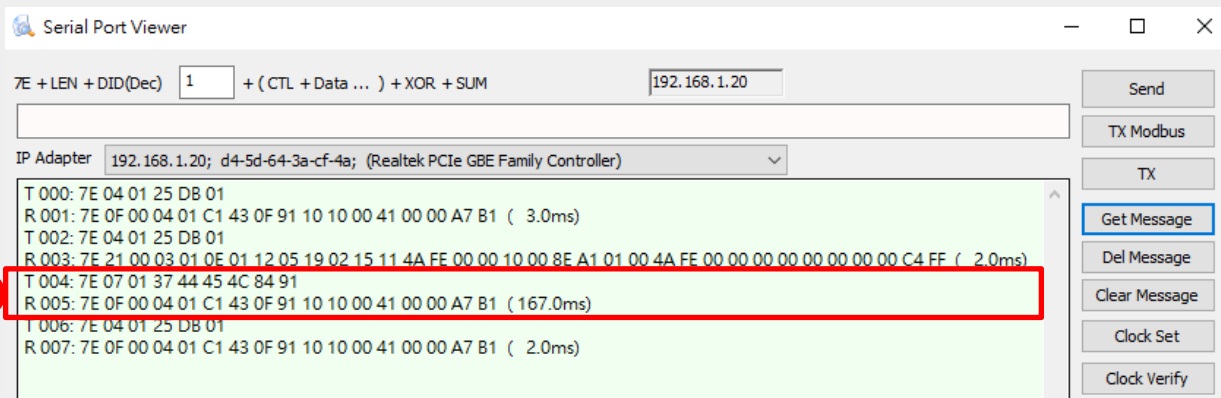
T 000: 7E 04 01 25 DB 01

R 001: 7E 0F 00 04 01 C1 43 0F 91 10 10 00 41 00 00 A7 B1 (3.0ms)



【25H】 / 【37H】

STEP 3 :刪除單筆歷史記錄



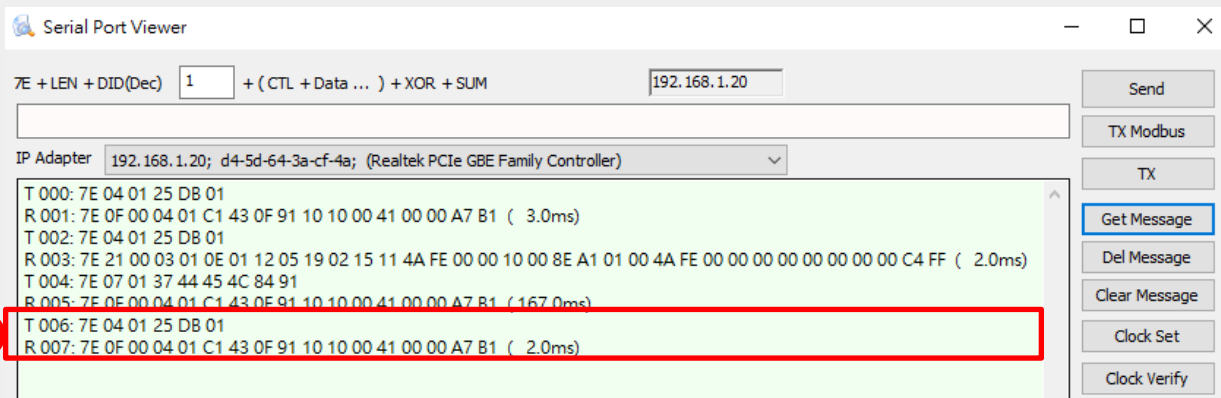
T 004: 7E 07 01 37 44 45 4C 84 91

R 005: 7E 0F 00 04 01 C1 43 0F 91 10 10 00 41 00 00 A7 B1 (167.0ms)



【25H】 / 【37H】

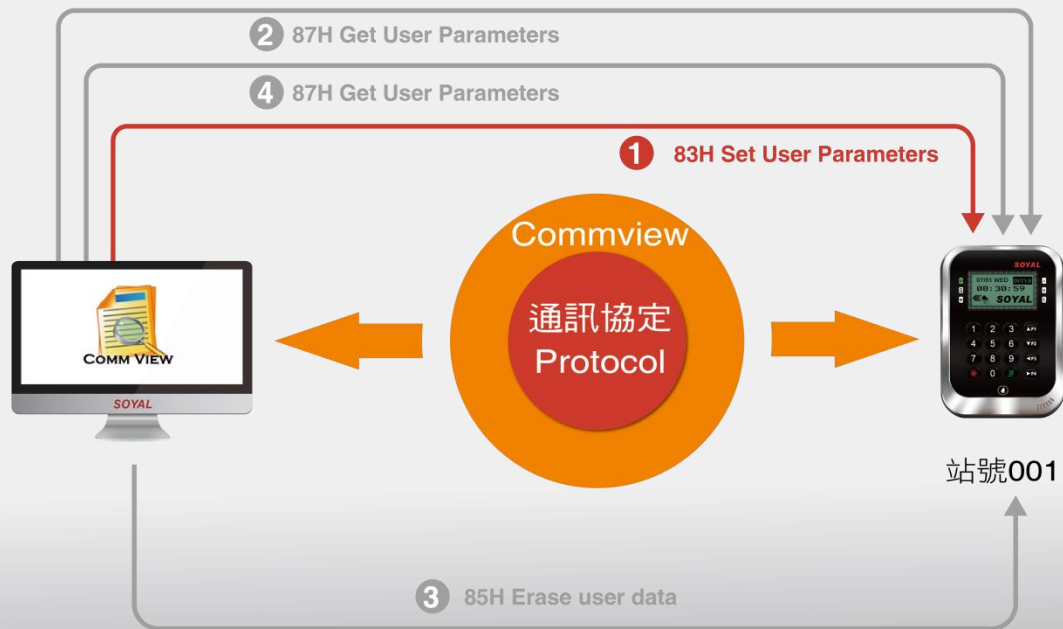
STEP 3 : 刪除卡片資料



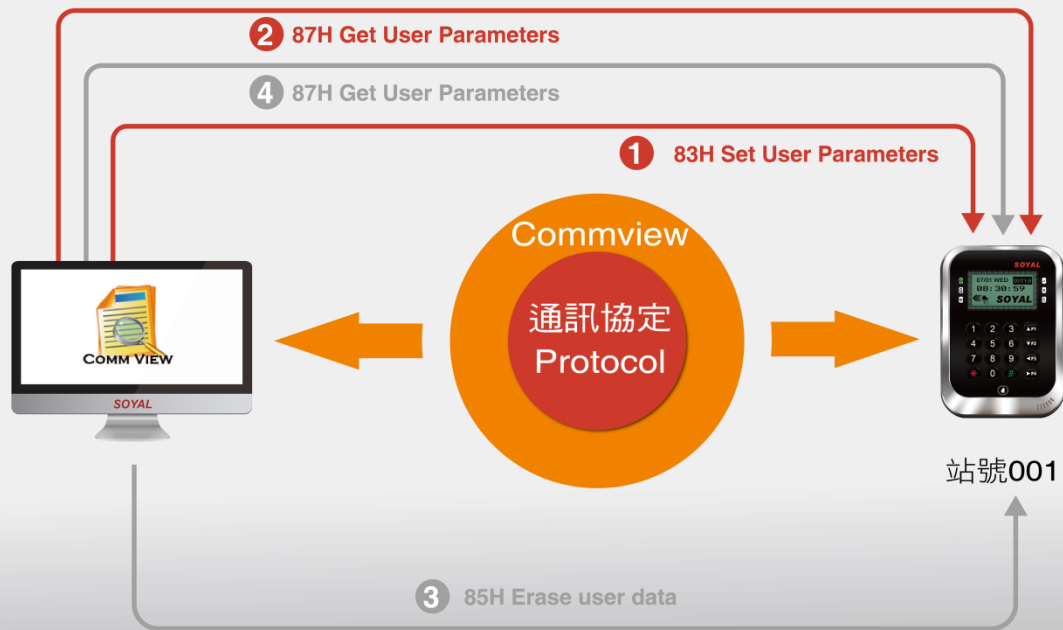
T 006: 7E 04 01 25 DB 01

R 007: 7E 0F 00 04 01 C1 43 0F 91 10 10 00 41 00 00 A7 B1 (2.0ms)

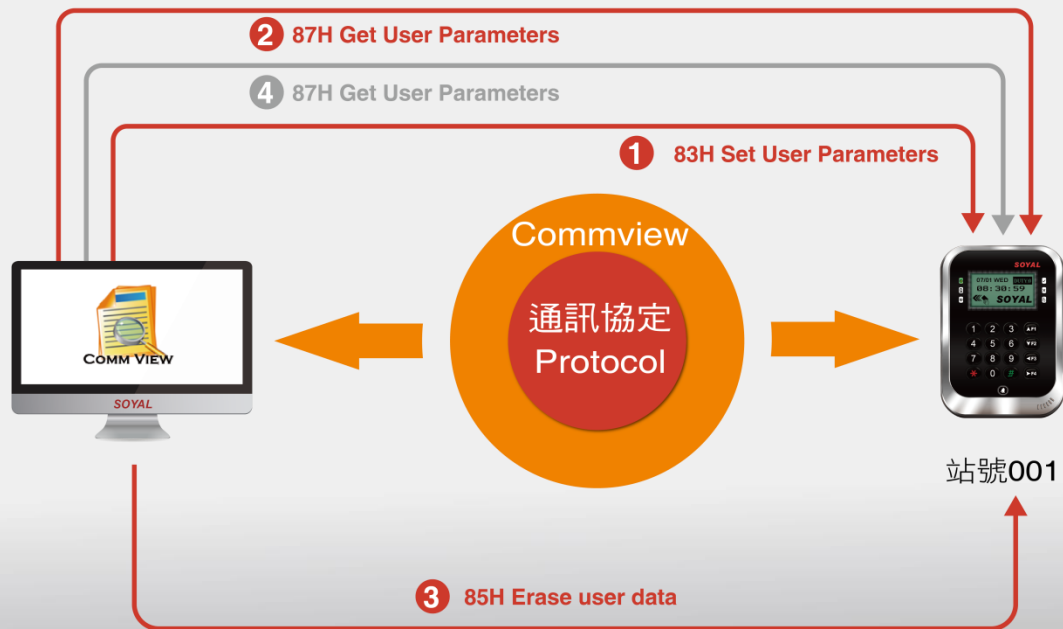
【83H/84H】 / 【85H】 / 【87H】 流程圖介紹



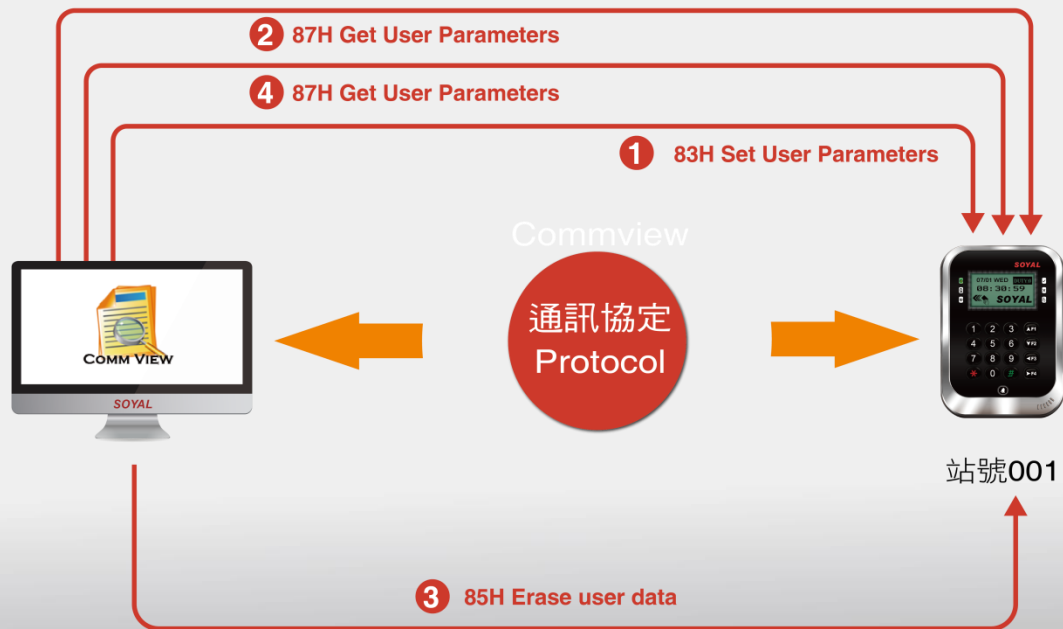
【83H/84H】 / 【85H】 / 【87H】 流程圖介紹



【83H/84H】 / 【85H】 / 【87H】 流程圖介紹



【83H/84H】 / 【85H】 / 【87H】 流程圖介紹



» H系列& E系列通訊協議差異

» 常用通訊協議指令概說

» 常用通訊協議指令實作



【83H/84H】Set User Parameters (家用版系列控制器)

Echo	Value	Description	
Head	7E	Leading Code	
Length	0E	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	83	Setting card content (AR-721H / AR-727HV3 only)	
Addr H	00	User Address – High	Addr H Addr L=0x01(00001) User Address : 00001
Addr L	01	User Address - Low	
Site H	04	Site Code – High	Site Hi Site Lo = 0x441 (01089) Site Code 01089
Site L	41	Site Code – Low	
Card H	EA	Card Code – High	Card Hi Card Lo = 0xEA4B (59979) Card Code 59979
Card L	4B	Card Code – Low	





【83H/84H】Set User Parameters (家用版系列控制器)

Echo	Value	Description	
PIN H	04	PIN – High	PIN H PIN L=0x4D2(1234) PIN 1234
PIN L	D2	PIN – Low	
Mode	02	Card or PIN	Access Mode(Bit 1~0)
			0 Invalid
			1 Read Only
			2 Card or PIN
			3 Card + PIN
		Bit7: Set to enable Anti passback	
Zone	01	Selecting the 1st Time Zone (11 Time Zones for selection)	
Group		Bit mapping of Door Group. (B0:WG1, B1:WG2, B3:WG3, B4:WG4) ※ for AR721E only.	
XOR	xx	$XOR = FF \wedge 01 \wedge 83 \wedge 00 \wedge 01 \wedge 04 \wedge 41 \wedge EA \wedge 4B \wedge 04 \wedge D2 \wedge 02 \wedge 01 = 4D$	
SUM	xx	$SUM = (01 + 83 + 00 + 01 + 04 + 41 + EA + 4B + 04 + D2 + 02 + 01 + 4D) \cdot FF = 25(LSB)$	

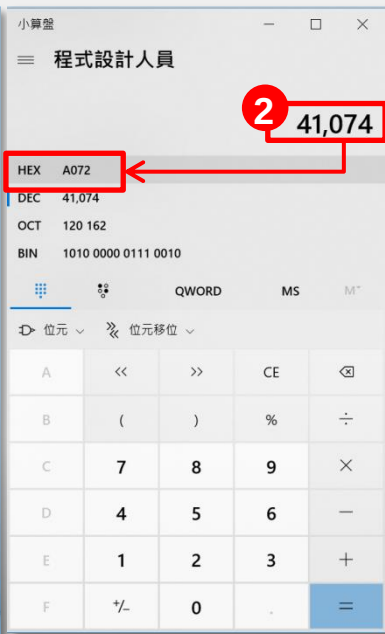
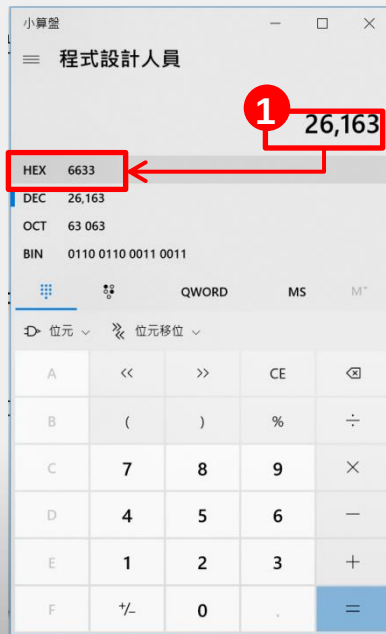
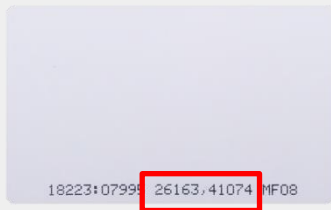


【83H/84H】Set User Parameters(家用版系列控制器)

Step 1: 輸入WG卡號前段
(Hex: 6633)

Step 2: 輸入WG卡號後段
(Hex: A072)

Step 3:
即可計算出完整HEX: 66 63 A0 72

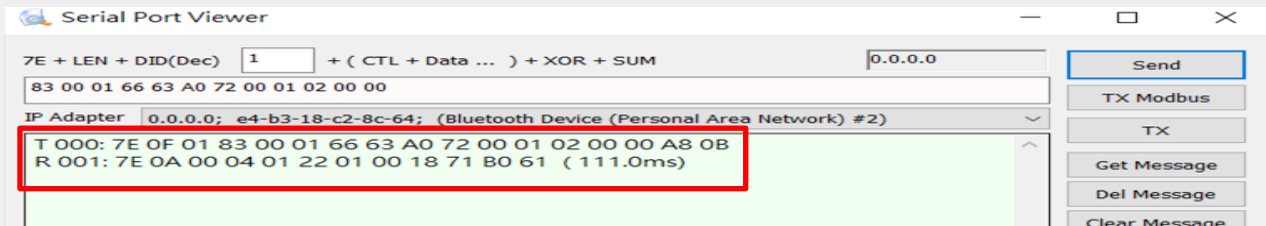




【83H/84H】Set User Parameters(家用版系列控制器)

家用版系列控制器的卡機下載單一張卡片

- 使用工具程式: Commview
- 卡號: 26163:41074
- Hex: 66 63 A0 72



T 000: 7E 0F 01 83 00 01 66 63 A0 72 00 01 02 00 00 A8 0B

R 001: 7E 0A 00 04 01 22 01 00 18 71 B0 61 (111.0ms)



【83H/84H】Set User Parameters (企業版系列控制器)

	Value	Description	
Head	??	Leading Code could be standard format or extended format	
Length	??	Data length can be one or two bytes depend on protocol format.	
DID	01	Node ID of destination, the node id is 01 here	
CMD	83/84	83H:with anti-passback , 84H:without anti-passback flag byte	
Records	??	How many records of user will be download to	
User Addr	??	Two bytes of user address. (Little endian)	
Tag UID	??	Eight bytes of Tag ID. (Little endian)	
PIN Code	??	Four bytes of PIN code. (Bit 31 ~ 24)	※3 Bitmask for HTZ 9~16
		Four bytes of PIN code. (Bit 23 ~ 16)	※3 Bitmask for HTZ 1 ~ 8
		Four bytes of PIN code. (Bit 15 ~ 8)	Word PIN Code (16 Bits)
		Four bytes of PIN code. (Bit 7 ~ 0)	



*另請參考指令2BH/8BH



【83H/84H】Set User Parameters (企業版系列控制器)

	Value	Description	
Mode	1	Bit7/6: Access Mode	Access Mode Define (Bit 7/6)
		Bit5 : patrol card	00 Invalid
		Bit4 : Card omitted after fingerprint recognition.	01 Read Only
		Bit3 : Fingerprint omitted after card recognition.	10 Card or PIN
		Bit2 : enable expire check	11 Card + PIN
		Bit1 : enable anti-pass back control	
		Bit0 : password change available	
Zone	1	Bit 7:	
		Multi door controller: Set to enable independent time zone for each node. (Cmd: 89H) Single door controller: Set to 0 if WG port will use the same time zone. Otherwise, the WG port will accord to Zone2. Bit 6: Reserved, must be 0. Bit5~0: User Access Time Zone. Zero for free zone control	



【83H/84H】Set User Parameters (企業版系列控制器)

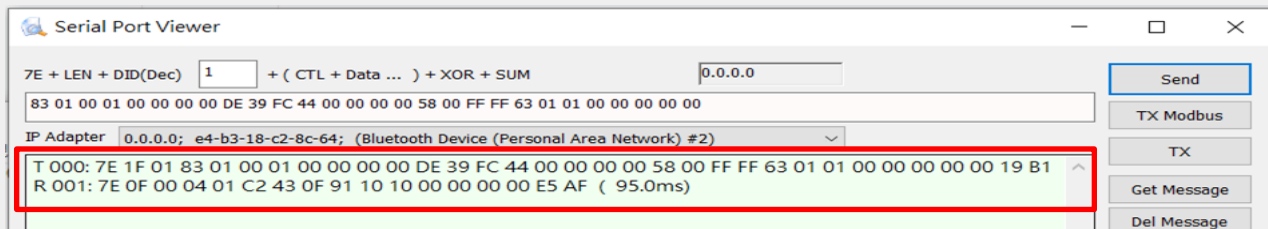
	Value	Description
Group 1	1	Bit mapping of available door. (Door 16~9/Bit 7~0) ※1
Group 2	1	Bit mapping of available door. (Door 8~1/Bit 7~0) ※1
Year	1	Year of the last allowed date at 2 digital format
Month	1	Month of the last allowed date at 2 digital format
Day	1	Day of the last allowed date at 2 digital format
Level	1	Bit7~6: User Level define range 0x00 to 0x03 Bit5~0: Zone 2 for WG port.
Option	1	Extend option Byte. Bit7: 1/0 En/Disable Anti-pass-back check Only available for cmd:83H, for 84H this byte will be discarded.
	3	Reserved
	N*26	Followed the next record of user data if records more than one set(Address UID PIN ...Reserved)
XOR	1	$XOR = FF \oplus DID \oplus CMD \oplus \dots \oplus \text{Level}$
SUM	1	$SUM = (DID + CMD + \dots + \text{Level} + XOR)$



【83H/84H】Set User Parameters (企業版系列控制器)

家用版系列控制器的卡機下載單一張卡片

- 使用工具程式: Commview
- 卡號: 56889:64580
- 用戶位置: 00001
- Hex: DE 39 FC 44



T 000: 7E 1F 01 83 01 00 01 00 00 00 00 DE 39 FC 44 00 00 00 00 58 00 FF FF 63 01 01 00 00 00 00 19 B1

R 001: 7E 0F 00 04 01 C2 43 0F 91 10 10 00 00 00 00 E5 AF (95.0ms)

» H系列& E系列通訊協議差異

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【87H】Get User Parameters

Echo	Value	Description	
Head	??	Leading Code could be standard format or extended format	
Length	07	Data length can be one or two bytes depend on protocol format.	
Node ID	01	Node ID of destination, the node id is 01 here	
Command	87	Get the user Parameters	
Addr H	00	User Address – High	Addr H Addr L=0x02(00002)
Addr L	02	User Address - Low	User Address : 00002
Nums	02	Continue number of cards	
XOR		XOR	
SUM		SUM	



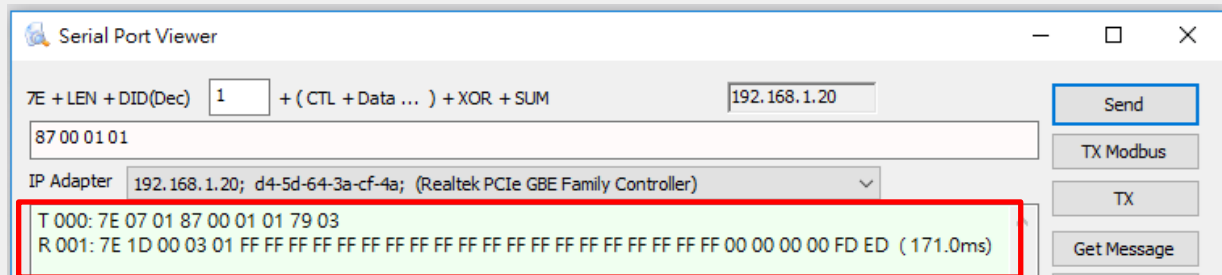
【87H】Get User Parameters

Echo	Value	Description	
Head	??	Leading Code could be standard format or extended format	
Length	??	Data length can be one or two bytes depend on protocol format.	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Function	03	Response to the request	
Source ID	??	The source node ID of data package come from (Who send)	
Data Field	24 bytes	The 1st User Parameters	Check reference to User Parameters Data Structure Please
	24 bytes	The 2nd User Parameters	
XOR			
SUM			



【87H】 Get User Parameters

- 使用工具程式: Commview
- 用戶位置: 00001



T 000: 7E 07 01 87 00 01 01 79 03

[illegible]

» H系列& E系列通訊協議差異

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» 常用通訊協議指令實作



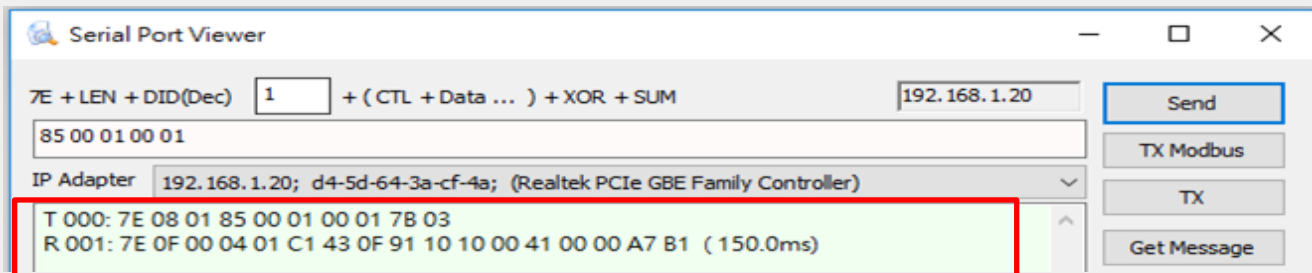
【85H】Erase user data (建議單筆刪除範圍:2000筆)

Echo	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	08	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination, the node id is 01 here
Command	85	Erase user data
Start-H		High byte of start user address (0 ~16383)
Start-L		Low byte of start user address
End-H		High byte of end(last) user address
End-L		Low byte of end(last) user address
XOR		
SUM		



【85H】Erase user data

- 使用工具程式: Commview
- 用戶位置: 00001



T 000: 7E 08 01 85 00 01 00 01 7B 03

R 001: 7E 0F 00 04 01 C1 43 0F 91 10 10 00 41 00 00 A7 B1 (150.0ms)

» H系列& E系列通訊協議差異

» 常用通訊協議指令概說

» 常用通訊協議指令實作



【21H】 Relay On/Off Control

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	21	Relay control (on or off), no indication for it
Data	82	Activating Relay (refer to 2.7.1)
XOR	5D	$XOR = FF \oplus 01 \oplus 21 \oplus 82 = 5D$
SUM	01	$SUM = (01 + 21 + 82 + 5D) \cdot FF = 01(LSB)$
Head	7E	Leading Code



【21H】Relay On/Off Control

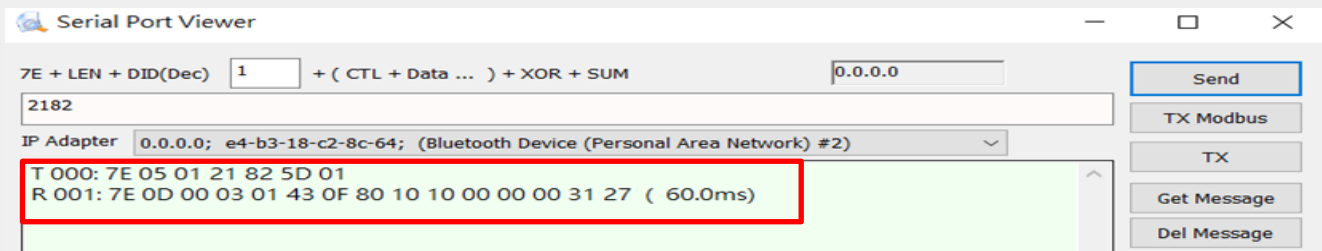
Relay Control Parameters

Data 0	Description	Note
0x00	Checking I/O Status	
0x01	Data1=(1/0)keypad lock/unlock	Only suits 6V2 or above
0x02	Data1/2=Interval between card flashing (about 10ms)	Only suits 6V2 or above
0x80	Activating Arming. (Enter armed state)	
0x81	Disable armed state	
0x82	Door Relay On	
0x83	Door Relay Off	
0x84	Activating Door Relay only for a certain time	* 1
0x85	Alarm Relay On	
0x86	Alarm Relay Off	
0x87	Activating Alarm Relay only for a certain time	* 1
0x88	On 721W Digital Output. 7E LEN DID 21 88 BIT	BIT2~0 : DO2~0
0x89	Off 721W Digital Output. 7E LEN DID 21 88 BIT	BIT2~0 : DO2~0



【21H】 Relay On/Off Control

- 使用工具程式: Commview



T 000: 7E 05 01 21 82 5D 01

R 001: 7E 0D 00 03 01 43 0F 80 10 10 00 00 00 31 27 (60.0ms)

3.常用通訊協議指令實作

Example 1 : Floor Download

Example 1 : Floor Download

- Read/Write User Floor (2FH) (Supported after version 2.09d and later)

Download/Write		
Echo	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	??	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination, the node id is 01 here
Command	2F	Set user access able floor data
Index HH	00	Start Index Bit 31~25.
Index HL	00	Start index Bit 24~16
Index LH	00	Start index Bit 15~08
Index LL	00	Start index Bit 07~00
Bytes		How many bytes of writing data. (Bit 15~08)
Bytes	N	How many bytes of writing data. (Bit 07~00)

Example 1 : Floor Download

- Read/Write User Floor (2FH) (Supported after version 2.09d and later)

Download/Write		
Echo	Value	Description
Data	xx	First data byte
...		
Data n		Last data byte
...		
XOR	xx	
SUM	xx	

Note:

If the index and records are all zero mean this command is erase all user data command.

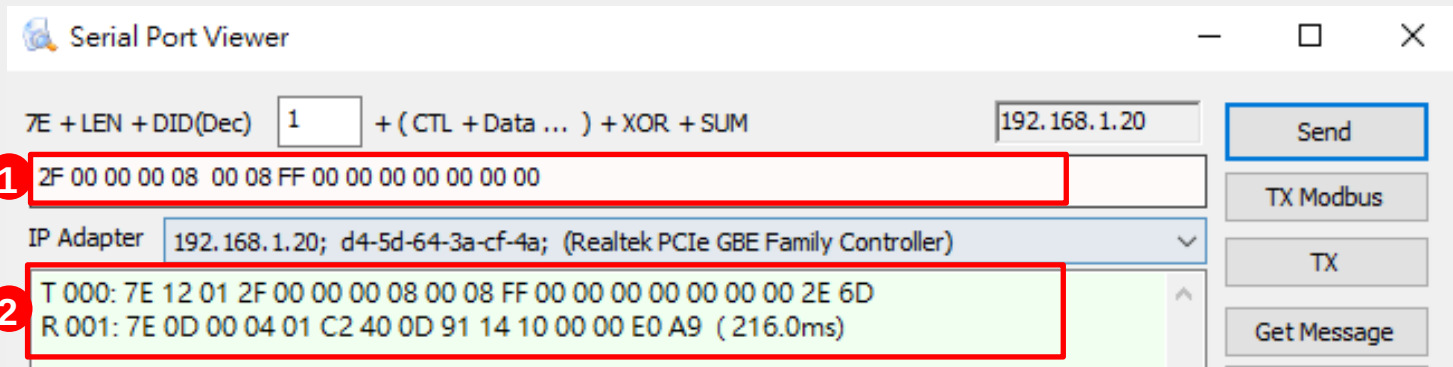
For speed up download purpose. We suggest you send erase command before download all data records.

Echo:

After command received the controller will echo ACK or NACK to the source controller or PC.

Example 1 : Floor Download

- 使用工具程式: Commview
- 指令 2FH
- 下載用戶位置 00001 樓層資料 1-8 點可以
- 用戶位置 00001



Step 1 : 寫入指令: 2F 00 00 00 08 00 08 FF 00 00 00 00 00 00 00



Step 2 : 出現以下指令

T 000: 7E 12 01 2F 00 00 00 08 00 08 FF 00 00 00 00 00 00 00 2E 6D

R 001: 7E 0D 00 04 01 C2 40 0D 91 14 10 00 00 E0 A9

Example 1 : Floor Download

輸入指令(T)

T 000: 7E 12 01 2F 00 00 00 08 00 08 FF 00 00 00 00 00 00 2E 6D

下載/上傳 樓層資料

固定值記錄的高低位元

樓層的對應位置

固定值 8Byte

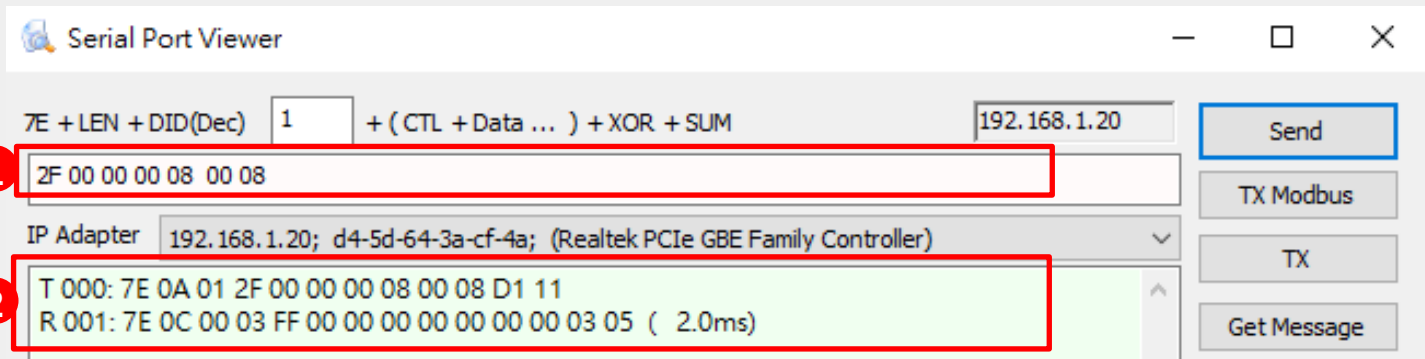
用戶位置 00001

回傳值(R)

R 001: 7E 0D 00 04 01 C2 40 0D 91 14 10 00 00 E0 A9

Example 1 : Floor Download

- 使用工具程式: Commview
- 讀取用戶位置 00001 樓層資料
- 指令 2FH
- 用戶位置 00001



Step 1 : 寫入指令: 2F 00 00 00 08 00 08

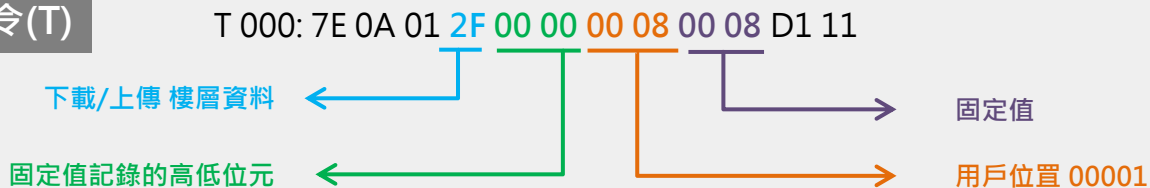
Step 2 : 出現以下指令

T 000: 7E 0A 01 2F 00 00 00 08 00 08 D1 11

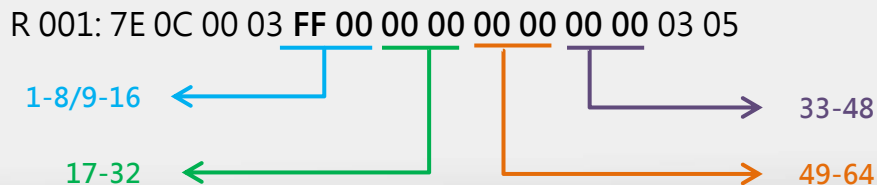
R 001: 7E 0C 00 03 FF 00 00 00 00 00 00 03 05

Example 1 : Floor Download

輸入指令(T)



回傳值(R)



Example 2 : DO On-Off

Example 2 : DO On-Off

- Set Output Status (17H)

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	xx	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination
Command	17	Set output status
Sub code		00h = Direct On/Off by Bitmap Table 01h = Latch On a specified output. 02h = Off a specified output. 03h = Flash a specified output at 1.67 Hz 04h = On a specified output with two bytes active interval on 10 time base. 05h = Flash On One Pp on 3Hz and limit total flash timer. 06h = Flash On One Pp on 3 Hz and limit pulse counter.
Parameters		Check the followed Parameters Table
...		
XOR		
SUM		

Example 2 : DO On-Off

- Parameters Table 1: Direct On/Off by Bitmap Table (00h):

P	Value	Description
Parameters	xx	Hi Byte of on Bit Table for output 15~8, Set Bit 0 to 1 will on output 8.
	xx	Lo byte of on Bit Table for output 7~0, Set Bit 0 to 1 will on output 0.
		Hi Byte of off Bit Table for output 15~8, Set Bit 0 to 1 will off output 8.
		Lo byte of off Bit Table for output 7~0, Set Bit 0 to 1 will off output 0.

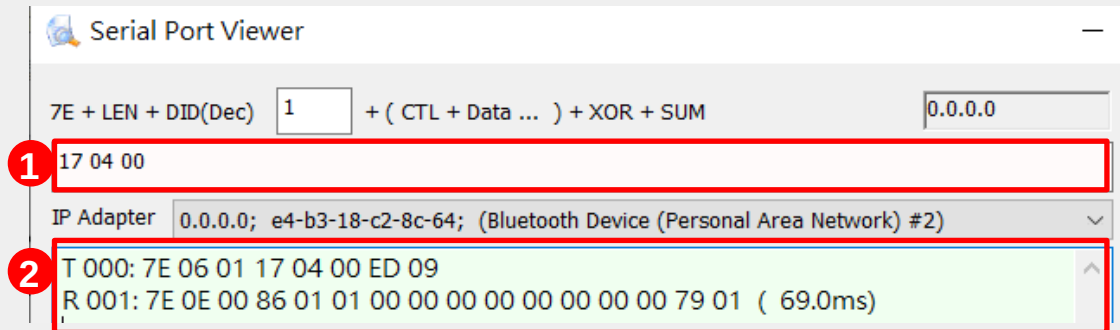
- Parameters 範例:

00= DO1 ON

01= DO2 ON

02= DO3 ON

03= DO4 ON

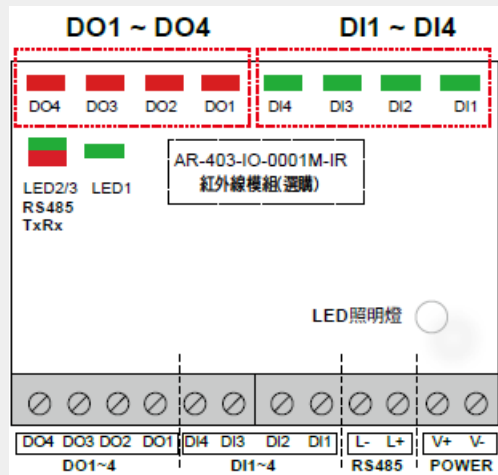
Example 2 : DO On-Off - DO On (以DO1為例)

Step 1 : 寫入指令: 17 04 00

Step 2 : 出現以下指令

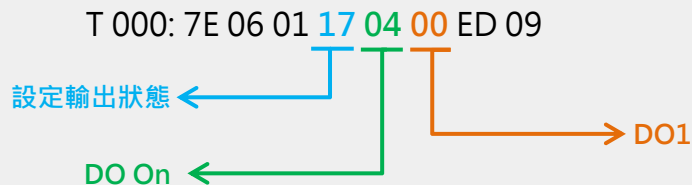
T 000: 7E 06 01 17 04 00 ED 09

R 001: 7E 0E 00 86 01 01 00 00 00 00 00 00 00 00 79 01



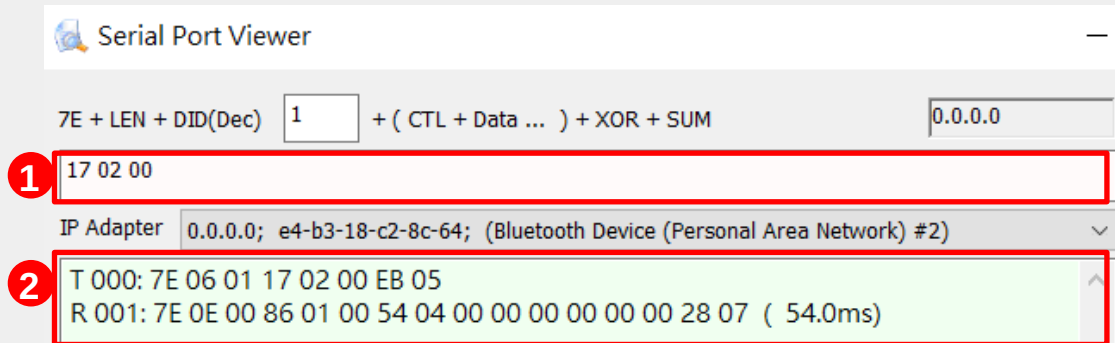
Example 2 : DO On-Off - DO On (以DO1為例)

輸入指令(T)



回傳值(R)

R 001: 7E 0E 00 86 01 01 00 00 00 00 00 00 00 00 79 01

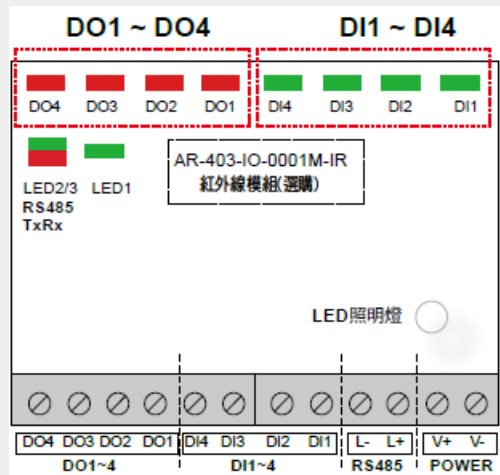
Example 2 : DO On-Off - DO Off (以DO1為例)

Step 1 : 寫入指令: 17 02 00

Step 2 : 出現以下指令

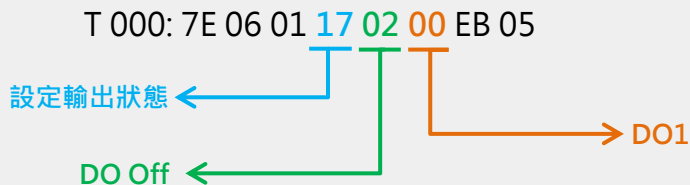
T 000: 7E 06 01 17 02 00 EB 05

R 001: 7E 0E 00 86 01 00 54 04 00 00 00 00 00 00 28 07



Example 2 : DO On-Off - **DO Off** (以DO1為例)

輸入指令(T)



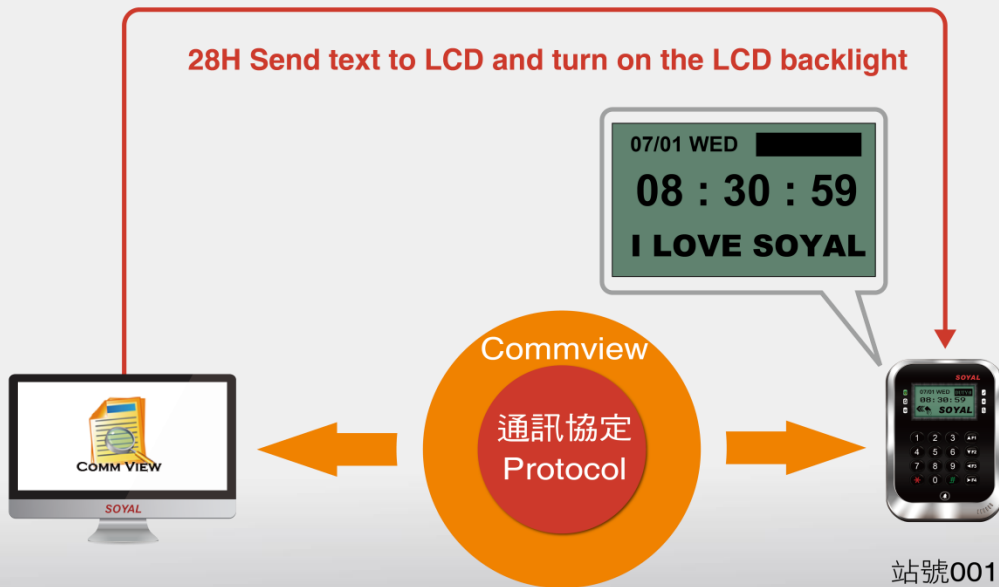
回傳值(R)

R 001: 7E 0E 00 86 01 01 00 00 00 00 00 00 00 00 79 01

Example 3 : 讀卡機 LCD 顯示控制

Example 3 : 讀卡機 LCD 顯示控制

- 設定螢幕顯示：I Love SOYAL



Example 3 : 讀卡機 LCD 顯示控制

- Send text to LCD and turn on the LCD backlight (28H)

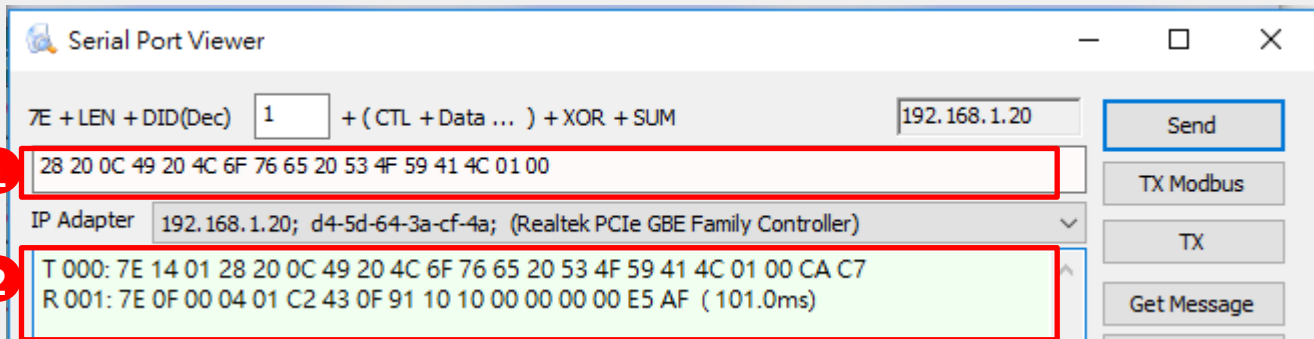
Echo	Value	Description
Head	7E	Leading Code could be standard format or extended format
Length	04	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination, the node id is 01 here
Command	28	Send text to LCD
	00 ~ 3F	LCD Column of first text. The index range must between 0x00 to 0x3F. 0x00 ~ 0x0F is first row. 0x10 ~ 0x1F is second row. 0x20 ~ 0x2F is third row. 0x30 ~ 0x3F is forth row.
	xx	How many bytes followed will be send to LCD
		[Text data bytes]
		Bit 15~08 of backlight delay time. (Base on 10ms)
		Bit 07~00 of backlight delay time. (Base on 10ms)
XOR	..	
SUM	..	

Echo:

After command received the controller will echo ACK or NACK to the source controller or PC.

P.S. Before send text to LCD need to lock LCD update via command 0x26 and release it after use.

Example 3 : 讀卡機 LCD 顯示控制



Step 1 : 寫入指令: 28 20 0C 49 20 4C 6F 76 65 20 53 4F 59 41 4C 01 00

Step 2 : 出現以下指令

T 000 : 7E 14 01 28 20 0C 49 20 4C 6F 76 65 20 53 4F 59 41 4C 01 00 CA C7

R 001 : 7E 0F 00 04 01 C2 43 0F 91 10 10 00 00 00 00 E5 AF (101.0ms)

Example 3 : 讀卡機 LCD 顯示控制

輸入指令(T)

T 000 : 7E 14 01 28 20 0C 49 20 4C 6F 76 65 20 53 4F 59 41 4C 01 00 CA C7

LCD顯示的行數，第三行
長度

I Love SOYAL

顯示停留LCD的秒數

回傳值(R)

R 001 : 7E 0F 00 04 01 C2 43 0F 91 10 10 00 00 00 00 E5 AF (101.0ms)

- I Love SOYAL 字串格式為Big 5，可使用以下查詢系統轉換：

BIG5碼線上查詢系統

Example 4 : 讀卡機 LCD/LED 及響聲控制

Example 4 :讀卡機 LCD/LED 及響聲控制

- 動作方式 :控制LCD 背光亮/紅/綠燈 亮



Example 4 :讀卡機 LCD/LED 及響聲控制

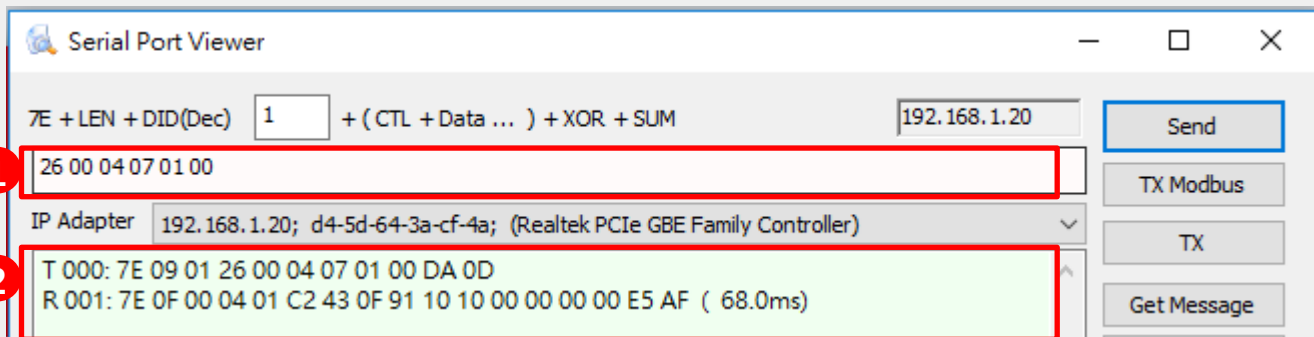
- Buzzer Sounds (26H)

Echo	Value	Description
Head	7E	Leading Code could be standard format or extended format
Length	05	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination, the node id is 01 here
Command	26	LED and Buzzer on/off control
	00 / 01	Channel assignment. 0x00 for controller and 0x01 for WG port
	xx	Beep will make (xx) sounds.
		Active LED/LCD assignment. Bit0: Set to on green LED lamp. Bit1: Set to on red LED lamp. Bit2: Set to On backlight LED. Bit3: Set to Off backlight LED. Bit4: Bit5: Bit6: Set to inhibit LCD update. (Reserved for user text via command 0x28) Bit7: Set to Lock The Keypad
		Bit 15~08 of LED delay time. (Base on 10ms),
		Bit 07~00 of LED delay time. (Base on 10ms). 0x0001 is Off command
XOR	..	
SUM	..	

Echo:

After command received the controller will echo ACK or NACK to the source controller or PC.

Example 4 :讀卡機 LCD/LED 及響聲控制



Step 1 : 寫入指令: 26 00 04 07 01 00

Step 2 : 出現以下指令

T 000: 7E 09 01 26 00 04 07 01 00 DA 0D

R 001: 7E 0F 00 04 01 C2 43 0F 91 10 10 00 00 00 00 E5 AF (68.0ms)

Example 4 :讀卡機 LCD/LED 及響聲控制

輸入指令(T)

T 000: 7E 09 01 26 00 04 07 01 00 DA 0D

0x00 for Controller

Beep 提示音 · 響4聲

LED 提示音的秒數

顯示背光白/紅/綠 燈 顯示

回傳值(R)

R 001: 7E 0F 00 04 01 C2 43 0F 91 10 10 00 00 00 00 E5 AF (68.0ms)

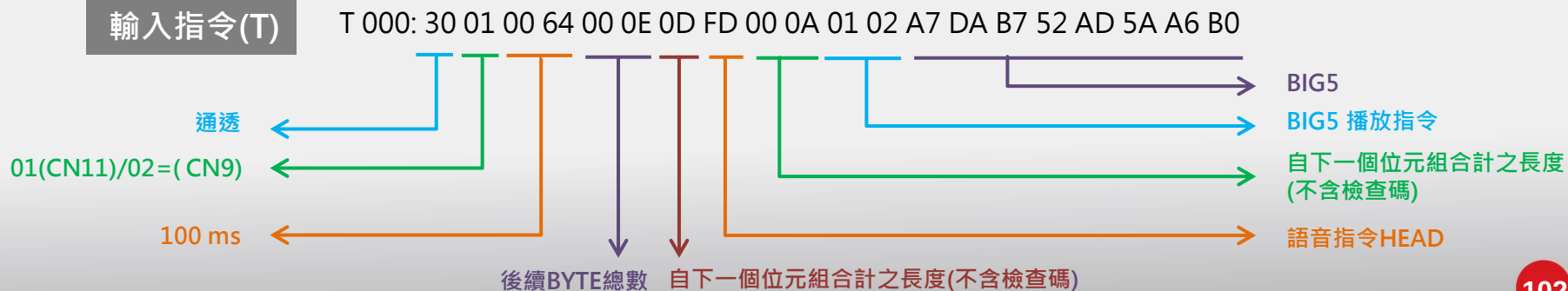
Example 5 :

使用語音功能說出 " 我愛茂旭 "

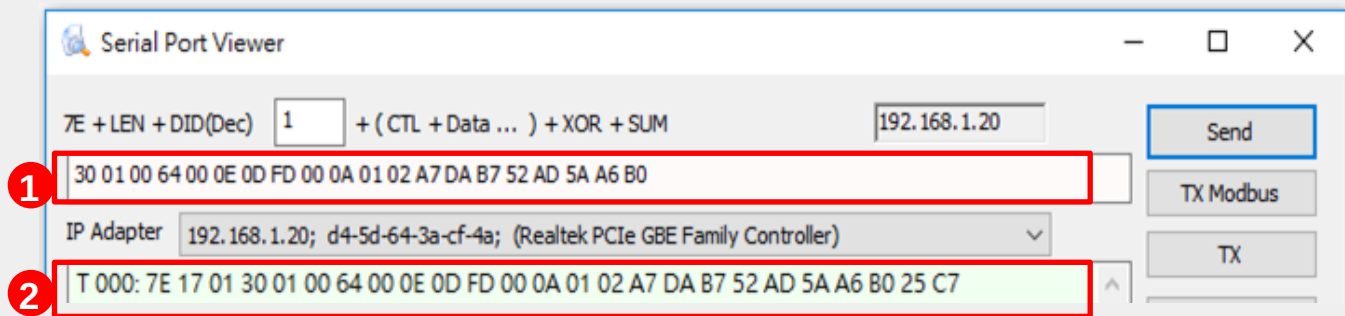
Example 4 : 使用語音功能說出 " 我愛茂旭 "

- DATA:資料說明

CMD	CN11	Time out	Bytes	0x000E Bytes							
30	01	00 64	00 0E	0D	FD	00 1A	01 02	A7 DA	B7 52	AD 5A	A6 B0
				Len	Voice	Len	Big 5	我	愛	茂	旭



Example 4 : 使用語音功能說出 “ 我愛茂旭 ”



Step 1 : 寫入指令: 30 01 00 64 00 0E 0D FD 00 0A 01 02 A7 DA B7 52 AD 5A A6 B0

Step 2 : 出現以下指令

T 000: 7E 17 01 30 01 00 64 00 0E 0D FD 00 0A 01 02 A7 DA B7 52 AD 5A A6 B0 25 C7